

A Systematic Survey of Model Predictive Path Integral Control: From Stochastic Theory to Real-Time Robotics

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ABSTRACT

Model Predictive Path Integral (MPPI) control, a sampling-based Stochastic Optimal Control (SOC) method, has achieved notable success in real-time robotics. However, its rapid proliferation of variants has produced a fragmented methodological landscape. This survey consolidates existing MPPI research by revisiting two complementary derivations within the SOC framework: the information-theoretic and the classical path integral formulations. A comparative analysis clarifies their mathematical equivalence under the noise-control matching condition and highlights their distinct assumptions. We then categorize existing approaches into four themes: sampling strategies, constraint handling, framework design, and robustness enhancement. We further review real-time implementations and present a modular, GPU-accelerated open-source MPPI framework, available at <https://anonymous.4open.science/r/mppi-framework/>. Finally, we outline open challenges and future directions, positioning this work as a research reference and technical roadmap.

1. Introduction

Model Predictive Control (MPC) has emerged as one of the most successful optimization-based control frameworks across robotics, autonomous driving, and process control applications [1]. By repeatedly solving finite-horizon optimization problems online, MPC enables systems to anticipate future states and optimize control sequences while explicitly handling constraints such as joint limits, actuator saturation, and collision avoidance. This predictive capability has proven effective in robotic manipulation [2], legged locomotion [3, 4], and autonomous navigation [5, 6], where complex task objectives and safety requirements must be systematically addressed.

However, traditional MPC approaches rely on gradient-based optimization under the critical assumption that both system dynamics and cost functions remain differentiable [7]. This assumption proves particularly limiting in robotics where systems frequently encounter non-smooth dynamics from contact events, discrete mode switches, and discontinuous cost landscapes. Moreover, the computational burden of derivative calculations scales poorly with system dimensionality, creating a fundamental tension between model fidelity and the stringent real-time requirements of robotic control, which can demand update rates as high as 50–1000 Hz. As a result, these limitations often force researchers and engineers to use simplified models, shorter planning horizons, or conservative approximations preventing MPC from realizing its theoretical advantages in complex robotic systems [8].

To address these limitations, sampling-based optimization techniques have emerged as an attractive alternative, circumventing the need for analytic gradients by evaluating multiple candidate trajectories in parallel. Pioneering works such as Stochastic Trajectory Optimization for Motion Planning (STOMP) [9] and Cross-Entropy Method (CEM) [10, 11] demonstrated the viability of this paradigm, particularly for handling non-differentiable cost functions and

leveraging Graphics Processing Unit (GPU) acceleration. However, STOMP primarily addressed open-loop trajectory optimization rather than closed-loop receding-horizon control, limiting its direct applicability to real-time robotic systems requiring continuous feedback [12]. While CEM can operate in a receding-horizon setting, its update rule is derived from cross-entropy minimization over elite samples rather than being directly derived from SOC framework.

These limitations motivated the development of Model Predictive Path Integral (MPPI) control [13], which combines a theoretically grounded update rule derived from the SOC framework with the receding-horizon structure of MPC. By expressing the optimal control as a weighted average over sampled noisy trajectories, MPPI sidesteps the need for explicit gradient computation while maintaining theoretical connections to optimal control. This combination of theoretical grounding and algorithmic simplicity has enabled efficient GPU-parallel implementations capable of handling complex, non-smooth cost landscapes found in many real-world control scenarios.

These properties have driven the adoption of MPPI across a broad spectrum of robotic and aerospace systems. In autonomous ground vehicles, MPPI has been deployed for both single-vehicle motion planning in cluttered urban environments [14] and decentralized multi-agent coordination, where prioritized variants such as P-MA-MPPI have reduced deadlocks and collisions while improving energy efficiency in autonomous mining haulage operations [15]. For robotic manipulation, MPPI has proven particularly effective for reactive, high-frequency control. Joint-space frameworks such as STORM [16] and the more recent single-instance sampling approach with dynamic time horizon [17] have pushed real-time control rates from 125 Hz to over 1 kHz on 7-DoF arms. Similar formulations have been extended to vision-based servoing [18], human-robot collaboration with learned pose prediction for anticipatory collision avoidance [19], and agricultural manipulation with deformable objects under Real2Sim2Real pipelines [20]. Aerial and aerospace systems have likewise leveraged the

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sampling-based nature of MPPI to handle nonlinear flight dynamics under environmental and actuator constraints, including agile quadrotor flight in cluttered environments at speeds up to 44 km/h [21], navigation under partial observability [22], and missile computational guidance with impact-angle, impact-time, field-of-view, and acceleration constraints [23, 24].

While these diverse applications demonstrate the versatility and effectiveness of MPPI, the rapid growth of MPPI research has resulted in a fragmented landscape of methodological contributions. Numerous variants have been proposed to address domain-specific challenges, including sampling efficiency enhancement techniques to reduce computational overhead, control barrier functions to enforce safety constraints, hierarchical decompositions for long-horizon planning, and robust formulations to handle model uncertainty, but these advances are scattered across the literature without systematic organization. As a result, the field lacks a clear overarching structure, making it difficult for researchers to gain a coherent understanding of existing approaches, identify common principles, and navigate the rapidly expanding body of work. This fragmentation also hinders the transfer of innovations between domains and complicates the selection of appropriate MPPI variants for new applications.

Recent efforts have begun to organize the rapidly expanding MPPI literature, most notably the surveys by Kazim *et al.* [25] and Honda [26]. Kazim *et al.* present a tutorial on the broader path integral control family, covering CEM, MPPI, and PI^2 -CMA along with their links to policy search and Reinforcement Learning (RL). Because this work spans the entire path integral family, MPPI is treated as one component among several, leaving its rapidly diversifying variants without a dedicated taxonomy. Their work also shows that the two derivations of MPPI yield the same value function, but the structural conditions and the correspondences between the objective functions, sampling distributions, and augmented cost terms are not examined. More recently, Honda recasts sampling-based MPC as probabilistic inference and reviews the literature along design dimensions such as prior shaping, multi-modality, and constraint handling. While this perspective unifies several algorithmic ideas under a common framework, the survey remains tutorial in nature, without comparative evaluation across common criteria or a reference implementation for reproducible benchmarking. In contrast to these prior efforts, which span the broader path integral family or adopt a probabilistic-inference framing, this paper presents a survey focused exclusively on MPPI from a stochastic optimal control perspective, with the following three key contributions:

- **Theoretical Foundations:** Two complementary derivations of MPPI from the information-theoretic and classical path integral formulations within the SOC framework. The comparative analysis establishes their mathematical equivalence under the noise-control

matching condition, examines the distinct assumptions of each derivation, and analyzes how MPPI relates to the open-loop, closed-loop, and deterministic formulations of optimal control.

- **Methodological Themes:** A systematic classification of MPPI research into four critical themes, namely sampling strategies, constraint handling, framework design, and robustness enhancement, is proposed to provide a coherent structure for the fragmented literature, accompanied by comparative analyses of representative variants under common criteria.
- **Open-Source Framework and Implementation:** An open-source, modular MPPI implementation with GPU-accelerated rollout engines and practical implementation guidelines, serving as both a reference baseline and a development platform for future research in robotics and control.

The remainder of this paper is organized as follows. Section 2 presents the theoretical foundations of MPPI and provides a comparative analysis of the two derivations. Section 3 systematically reviews methodological advances across four key themes. Section 4 reviews the evolution of computational implementations and introduces a modular open-source framework. Section 5 concludes with open challenges and future research directions.

2. Theoretical Foundations of MPPI

This section presents the theoretical foundations of MPPI control through two complementary derivations within the SOC framework. Although the two derivations start from fundamentally different mathematical machinery, information-theoretic divergence minimization and the continuous-time HJB equation, both arrive at the same importance-weighted control update, where sampled trajectories are weighted by exponentiated costs and averaged to produce the control input. The comparative analysis that follows explains why this convergence occurs under specific structural conditions, where the two derivations diverge in their assumptions, and what this implies for the optimality and applicability of MPPI.

2.1. Information-Theoretic Derivation of MPPI

This subsection provides a concise derivation of MPPI from an information-theoretic perspective, following the framework established by Williams *et al.* [27]. For a more comprehensive treatment of the theoretical details, readers are referred to the original work.

Consider a discrete-time dynamical system

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{v}_t), \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{u}_t, \Sigma), \quad (1)$$

where $\mathbf{x}_t \in \mathbb{R}^n$ is the state vector and $\mathbf{v}_t \in \mathbb{R}^m$ is a noisy control input drawn from a Gaussian centered at the controllable mean $\mathbf{u}_t \in \mathbb{R}^m$. Let $\mathbf{U} = (\mathbf{u}_0, \dots, \mathbf{u}_{T-1}) \in \mathbb{R}^{m \times T}$ denote the sequence of controllable mean inputs, and

$\mathbf{V} = (\mathbf{v}_0, \dots, \mathbf{v}_{T-1}) \in \mathbb{R}^{m \times T}$ denote the corresponding noisy control sequence. The probability density of $\mathbf{V}$ given $\mathbf{U}$ is given by

$$q(\mathbf{V} | \mathbf{U}, \Sigma) = \prod_{t=0}^{T-1} Z^{-1} \exp\left(-\frac{1}{2}(\mathbf{v}_t - \mathbf{u}_t)^\top \Sigma^{-1}(\mathbf{v}_t - \mathbf{u}_t)\right), \quad (2)$$

where $Z = (2\pi)^{\frac{m}{2}} |\Sigma|^{\frac{1}{2}}$ is the normalization constant of the Gaussian density. Note that although we present the derivation with a constant Σ for notational simplicity, the formulation extends directly to the time-varying case Σ_t , as the per-step factorization in (2) does not require identical covariances across time steps. We denote by $\mathbb{Q}$ the probability distribution whose density is given by $q(\mathbf{V} | \mathbf{U}, \Sigma)$. Given a control sequence $\mathbf{V}$, the resulting state trajectory $\{\mathbf{x}_t\}_{t=0}^T$ is determined by the system dynamics (1).

The state-dependent cost is defined as

$$S(\mathbf{V}) = \phi(\mathbf{x}_T) + \sum_{t=0}^{T-1} \ell(\mathbf{x}_t), \quad (3)$$

where $\phi(\mathbf{x}_T)$ is the terminal state cost and $\ell(\mathbf{x}_t)$ is the stage state cost at time t . Using this cost definition, the SOC problem can be formulated to establish the connection with information theory, as follows:

$$\mathbf{U}^* = \arg \min_{\mathbf{U}} \mathbb{E}_{\mathbb{Q}} \left[S(\mathbf{V}) + \frac{\lambda}{2} \sum_{t=0}^{T-1} (\mathbf{u}_t^\top \Sigma^{-1} \mathbf{u}_t + \boldsymbol{\beta}_t^\top \mathbf{u}_t) \right], \quad (4)$$

where $\boldsymbol{\beta}_t \in \mathbb{R}^m$ is the coefficient vector of the linear control cost term, and $\lambda > 0$ is a weighting parameter that scales the quadratic control cost.

To enable the information-theoretic derivation, we introduce the free-energy functional:

$$\mathcal{F}_\lambda = -\lambda \log \mathbb{E}_{\mathbb{P}} \left[\exp\left(-\frac{1}{\lambda} S(\mathbf{V})\right) \right], \quad (5)$$

where $\mathbb{P}$ represents the base distribution over control sequences $\mathbf{V}$, typically chosen as zero-mean Gaussian noise: $\mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \Sigma)$. Unlike in (4), where λ appeared as a weighting factor for the quadratic control cost, here λ serves as a temperature parameter. To derive an upper bound on (5), we introduce a controlled distribution $\mathbb{Q}$ over control sequences and rewrite the expectation in (5) via importance sampling:

$$\mathcal{F}_\lambda = -\lambda \log \mathbb{E}_{\mathbb{Q}} \left[\exp\left(-\frac{1}{\lambda} S(\mathbf{V})\right) \frac{p(\mathbf{V})}{q(\mathbf{V})} \right]. \quad (6)$$

Since the logarithm is concave, applying Jensen's inequality to (6) yields

$$\mathcal{F}_\lambda \leq -\lambda \mathbb{E}_{\mathbb{Q}} \left[\log \left(\exp\left(-\frac{1}{\lambda} S(\mathbf{V})\right) \frac{p(\mathbf{V})}{q(\mathbf{V})} \right) \right]. \quad (7)$$

Expanding the logarithm in (7) and using the definition of the KL divergence gives

$$\mathcal{F}_\lambda \leq \mathbb{E}_{\mathbb{Q}} [S(\mathbf{V})] + \lambda \mathbb{D}_{\text{KL}}(\mathbb{Q} \| \mathbb{P}). \quad (8)$$

To establish the connection with (4), we set the base distribution as $\mathbb{P} = q(\mathbf{V} | \tilde{\mathbf{U}}, \Sigma)$ and the controlled distribution as $\mathbb{Q} = q(\mathbf{V} | \mathbf{U}, \Sigma)$, where $\tilde{\mathbf{U}}$ represents a reference control sequence. Since both $\mathbb{P}$ and $\mathbb{Q}$ are Gaussian with identical covariance Σ and different means, the KL divergence term can be written in closed form as

$$\mathbb{D}_{\text{KL}}(\mathbb{Q} \| \mathbb{P}) = \frac{1}{2} \sum_{t=0}^{T-1} (\mathbf{u}_t - \tilde{\mathbf{u}}_t)^\top \Sigma^{-1} (\mathbf{u}_t - \tilde{\mathbf{u}}_t). \quad (9)$$

Expanding the quadratic term in (9) yields

$$\mathbb{D}_{\text{KL}}(\mathbb{Q} \| \mathbb{P}) = \frac{1}{2} \sum_{t=0}^{T-1} (\mathbf{u}_t^\top \Sigma^{-1} \mathbf{u}_t - 2\tilde{\mathbf{u}}_t^\top \Sigma^{-1} \mathbf{u}_t + \tilde{\mathbf{u}}_t^\top \Sigma^{-1} \tilde{\mathbf{u}}_t). \quad (10)$$

Substituting (10) into (8), and using the definition of $S(\mathbf{V})$ in (3), yields

$$\mathcal{F}_\lambda \leq \mathbb{E}_{\mathbb{Q}} \left[\phi(\mathbf{x}_T) + \sum_{t=0}^{T-1} (\ell(\mathbf{x}_t) + \frac{\lambda}{2} (\mathbf{u}_t^\top \Sigma^{-1} \mathbf{u}_t + \boldsymbol{\beta}_t^\top \mathbf{u}_t + c_t)) \right] \quad (11)$$

where $\boldsymbol{\beta}_t := -2 \Sigma^{-1} \tilde{\mathbf{u}}_t$, $c_t := \tilde{\mathbf{u}}_t^\top \Sigma^{-1} \tilde{\mathbf{u}}_t$ represent the linear and constant components resulting from shifting the mean control from $\tilde{\mathbf{U}}$ to $\mathbf{U}$.

As a result, the cost structure in (11) aligns with that in (4), implying that the free energy serves as a lower bound on the optimal control objective. The bound in (8) is achieved with equality for a specific distribution, which we define as the optimal distribution $\mathbb{Q}^*$ with density

$$q^*(\mathbf{V}) = \frac{1}{\eta} \exp\left(-\frac{1}{\lambda} S(\mathbf{V})\right) p(\mathbf{V}), \quad (12)$$

where

$$\eta = \int \exp\left(-\frac{1}{\lambda} S(\mathbf{V})\right) p(\mathbf{V}) d\mathbf{V} \quad (13)$$

is the normalizing constant. This can be verified by substituting (12) into the KL divergence in (8), which causes the expected cost terms to cancel and reduces the right-hand side to $\mathcal{F}_\lambda$ itself [27].

Using this definition, the SOC problem in (4) can be reformulated as minimizing the KL divergence between the optimal distribution $\mathbb{Q}^*$ and the controlled distribution $\mathbb{Q}$:

$$\begin{aligned} \mathbf{U}^* &= \arg \min_{\mathbf{U}} \mathbb{D}_{\text{KL}}(\mathbb{Q}^* \| \mathbb{Q}) \\ &= \arg \min_{\mathbf{U}} \mathbb{E}_{\mathbb{Q}^*} \left[\sum_{t=0}^{T-1} (\mathbf{v}_t - \mathbf{u}_t)^\top \Sigma^{-1} (\mathbf{v}_t - \mathbf{u}_t) \right] \\ &= \mathbb{E}_{\mathbb{Q}^*} [\mathbf{V}]. \end{aligned} \quad (14)$$

Since direct sampling from the optimal distribution $\mathbb{Q}^*$ is intractable, MPPI employs importance sampling using the Gaussian distribution $q(\mathbf{V} | \hat{\mathbf{U}}, \Sigma)$, where $\hat{\mathbf{U}}$ represents the

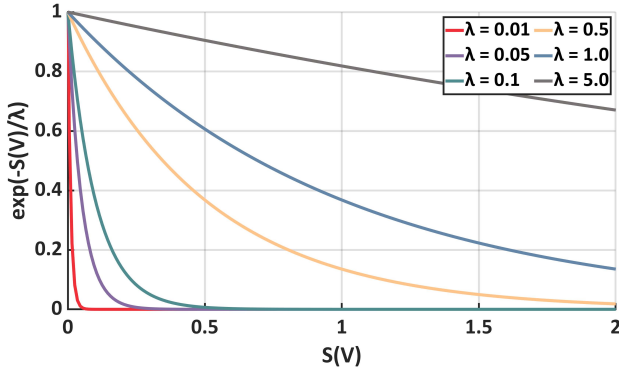


Figure 1: Effect of the temperature parameter λ on the exponential weighting function $\exp(-S(V)/\lambda)$. Smaller λ values yield sharper decay, emphasizing low-cost trajectories, whereas larger λ values produce smoother weighting and higher exploration.

current control estimate. The optimization in (14) can then be reformulated as follows:

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}^*}[\mathbf{V}] &= \int q^*(\mathbf{V}) \mathbf{V} d\mathbf{V} \\ &= \int \frac{q^*(\mathbf{V})}{q(\mathbf{V} | \hat{\mathbf{U}}, \Sigma)} q(\mathbf{V} | \hat{\mathbf{U}}, \Sigma) \mathbf{V} d\mathbf{V} \quad (15) \\ &= \mathbb{E}_{\mathbb{Q}}[w(\mathbf{V}) \mathbf{V}]. \end{aligned}$$

The importance weight $w(\mathbf{V})$ is derived by factorizing through the base distribution $p(\mathbf{V})$:

$$\begin{aligned} w(\mathbf{V}) &= \left(\frac{q^*(\mathbf{V})}{p(\mathbf{V})} \right) \left(\frac{p(\mathbf{V})}{q(\mathbf{V} | \hat{\mathbf{U}}, \Sigma)} \right) \\ &= \frac{1}{\eta} \exp\left(-\frac{1}{\lambda} S(\mathbf{V})\right) \frac{p(\mathbf{V})}{q(\mathbf{V} | \hat{\mathbf{U}}, \Sigma)} \quad (16) \\ &= \frac{1}{\eta} \exp\left(-\frac{1}{\lambda} (S(\mathbf{V}) + \lambda \sum_{i=0}^{T-1} (\hat{\mathbf{u}}_i - \tilde{\mathbf{u}}_i)^\top \Sigma^{-1} \mathbf{v}_i)\right). \end{aligned}$$

The importance weight in (16) is exponentially shaped by the temperature parameter λ , which controls the concentration of the sampling distribution around low-cost trajectories, as illustrated in Fig. 1. Substituting the importance weight derived in (16) into (15) yields the theoretical MPPI control law. Although this solution is theoretically global, it requires evaluating expectations over the entire trajectory space, which is computationally intractable.

In practice, this expectation is approximated via Monte Carlo (MC) sampling with K rollout trajectories, yielding the normalized importance weights:

$$w^{(k)} = \frac{\exp\left(-\frac{1}{\lambda} (S(\mathbf{V}^{(k)}) + \lambda \sum_{i=0}^{T-1} (\hat{\mathbf{u}}_i - \tilde{\mathbf{u}}_i)^\top \Sigma^{-1} \mathbf{v}_i^{(k)})\right)}{\sum_{j=1}^K \exp\left(-\frac{1}{\lambda} (S(\mathbf{V}^{(j)}) + \lambda \sum_{i=0}^{T-1} (\hat{\mathbf{u}}_i - \tilde{\mathbf{u}}_i)^\top \Sigma^{-1} \mathbf{v}_i^{(j)})\right)}. \quad (17)$$

The control update is then obtained as the weighted average of the sampled perturbations:

$$\mathbf{u}_t = \hat{\mathbf{u}}_t + \sum_{k=1}^K w^{(k)} \delta \mathbf{u}_t^{(k)}, \quad \delta \mathbf{u}_t^{(k)} \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad (18)$$

where $\delta \mathbf{u}_t^{(k)} = \mathbf{v}_t^{(k)} - \hat{\mathbf{u}}_t$ denotes the control perturbation for the k -th sample. This MC approximation constitutes the practical MPPI algorithm and forms the foundation for real-time implementations.

2.2. Classical Path Integral Derivation of MPPI

This subsection presents the derivation of MPPI from the perspective of classical path integral control theory. The key insight is transforming the nonlinear HJB equation into a linear partial differential equation through an exponential transformation of the value function, enabling forward sampling of stochastic trajectories rather than backward value function computation. This linearization technique was originally introduced by Kappen [28, 29] for a class of nonlinear stochastic control problems with control-affine dynamics and quadratic control costs, and was later extended to the MPPI setting by Williams *et al.* [30]. For readers interested in the detailed mathematical foundations of this classical path integral formulation, we refer to [28, 29] and [30].

We begin by formulating a general continuous-time stochastic dynamical system as follows:

$$d\mathbf{x}_t = f(\mathbf{x}_t) dt + G(\mathbf{x}_t) \mathbf{u}_t dt + B(\mathbf{x}_t) d\mathbf{w}_t, \quad (19)$$

where $G(\mathbf{x}_t) \in \mathbb{R}^{n \times m}$ denotes the state-dependent control transition matrix, $B(\mathbf{x}_t) \in \mathbb{R}^{n \times q}$ represents the diffusion matrix, and $d\mathbf{w}_t \in \mathbb{R}^q$ is the differential of a standard Brownian motion capturing stochastic disturbances. The state is partitioned into a passive part and a controlled part, and the system components $\mathbf{x}$, f , G , B are expressed as follows:

$$\begin{aligned} \mathbf{x}_t &= \begin{bmatrix} \mathbf{x}_t^p \\ \mathbf{x}_t^c \end{bmatrix}, \quad f(\mathbf{x}_t) = \begin{bmatrix} f_p(\mathbf{x}_t) \\ f_c(\mathbf{x}_t) \end{bmatrix}, \\ G(\mathbf{x}_t) &= \begin{bmatrix} \mathbf{0}_{\ell \times m} \\ G_c(\mathbf{x}_t) \end{bmatrix}, \quad B(\mathbf{x}_t) = \begin{bmatrix} \mathbf{0}_{\ell \times q} \\ B_c(\mathbf{x}_t) \end{bmatrix}. \end{aligned} \quad (20)$$

Only the controlled part $\mathbf{x}_t^c$ is directly influenced by control inputs and process noise, whereas the passive part $\mathbf{x}_t^p$ is not directly actuated and has no direct diffusion term.

To derive the path integral formulation, let us first establish the classical stochastic optimal control framework. The value function, which represents the optimal expected future cost from state $\mathbf{x}_t$ at time t , is defined as

$$V(\mathbf{x}_t, t) = \min_{\mathbf{u}(\cdot)} \mathbb{E}_{\mathbb{Q}} \left[\phi(\mathbf{x}_T) + \int_t^T \left(\ell(\mathbf{x}_\tau) + \frac{1}{2} \mathbf{u}_\tau^\top R \mathbf{u}_\tau \right) d\tau \right], \quad (21)$$

where $R \in \mathbb{R}^{m \times m}$ is positive definite. Given the stochastic optimal control problem defined by (19)–(21), the value

function $V(\mathbf{x}_t, t)$ satisfies the stochastic HJB equation

$$-\partial_t V(\mathbf{x}_t, t) = \min_{\mathbf{u}_t} \left[\ell(\mathbf{x}_t) + \frac{1}{2} \mathbf{u}_t^\top \mathbf{R} \mathbf{u}_t + \left(f(\mathbf{x}_t) + \mathbf{G}(\mathbf{x}_t) \mathbf{u}_t \right)^\top \nabla_{\mathbf{x}} V(\mathbf{x}_t, t) + \frac{1}{2} \text{tr} \left(\mathbf{B}(\mathbf{x}_t) \mathbf{B}(\mathbf{x}_t)^\top \nabla_{\mathbf{x}\mathbf{x}}^2 V(\mathbf{x}_t, t) \right) \right]. \quad (22)$$

The control-dependent terms in (22) are

$$\frac{1}{2} \mathbf{u}_t^\top \mathbf{R} \mathbf{u}_t + \mathbf{u}_t^\top \mathbf{G}(\mathbf{x}_t)^\top \nabla_{\mathbf{x}} V(\mathbf{x}_t, t). \quad (23)$$

Since these terms are quadratic in $\mathbf{u}_t$, the minimizer can be obtained analytically. Taking the gradient of (23) with respect to $\mathbf{u}_t$ and setting it to zero gives

$$\mathbf{R} \mathbf{u}_t + \mathbf{G}(\mathbf{x}_t)^\top \nabla_{\mathbf{x}} V(\mathbf{x}_t, t) = 0. \quad (24)$$

Therefore, the optimal control law is

$$\mathbf{u}^*(\mathbf{x}_t, t) = -\mathbf{R}^{-1} \mathbf{G}(\mathbf{x}_t)^\top \nabla_{\mathbf{x}} V(\mathbf{x}_t, t). \quad (25)$$

Substituting (25) into (22) eliminates the minimization operator and yields the reduced nonlinear HJB equation

$$-\partial_t V(\mathbf{x}_t, t) = \ell(\mathbf{x}_t) + f(\mathbf{x}_t)^\top \nabla_{\mathbf{x}} V(\mathbf{x}_t, t) - \frac{1}{2} \nabla_{\mathbf{x}} V(\mathbf{x}_t, t)^\top \mathbf{G}(\mathbf{x}_t) \mathbf{R}^{-1} \mathbf{G}(\mathbf{x}_t)^\top \nabla_{\mathbf{x}} V(\mathbf{x}_t, t) + \frac{1}{2} \text{tr} \left(\mathbf{B}(\mathbf{x}_t) \mathbf{B}(\mathbf{x}_t)^\top \nabla_{\mathbf{x}\mathbf{x}}^2 V(\mathbf{x}_t, t) \right), \quad (26)$$

with the boundary condition,

$$V(\mathbf{x}_T, T) = \phi(\mathbf{x}_T). \quad (27)$$

This reduced nonlinear HJB equation is equivalent to the stochastic HJB equation and characterizes the same optimal value function.

Although the HJB equation (26) characterizes the optimal value function, it must be solved backward in time from the terminal state to the initial state. This backward propagation requires evaluating the value function over the entire state-time space, which quickly becomes computationally intractable as the state dimension increases, commonly referred to as the curse of dimensionality. To overcome this issue, the nonlinear HJB equation can be transformed into a linear partial differential equation, allowing the use of forward sampling of stochastic trajectories. This transformation can be achieved by applying an exponential transformation to the value function. Specifically, the desirability function $\Psi(\mathbf{x}_t, t)$ is defined as:

$$V(\mathbf{x}_t, t) = -\lambda \log \Psi(\mathbf{x}_t, t). \quad (28)$$

By differentiating (28) with respect to t and $\mathbf{x}_t$ and by computing the Hessian, the following expressions are obtained:

$$\begin{aligned} \partial_t V &= -\frac{\lambda}{\Psi} \frac{\partial \Psi}{\partial t}, \quad \nabla_{\mathbf{x}} V = -\frac{\lambda}{\Psi} \nabla_{\mathbf{x}} \Psi, \\ \nabla_{\mathbf{x}\mathbf{x}}^2 V &= -\frac{\lambda}{\Psi} \nabla_{\mathbf{x}\mathbf{x}}^2 \Psi + \frac{\lambda}{\Psi^2} \nabla_{\mathbf{x}} \Psi \nabla_{\mathbf{x}} \Psi^\top. \end{aligned} \quad (29)$$

Substituting (29) into the HJB equation (26), the equation can be rewritten as a PDE with respect to the desirability function $\Psi(\mathbf{x}_t, t)$ as follows:

$$\begin{aligned} \partial_t \Psi(\mathbf{x}_t, t) &= \frac{\Psi(\mathbf{x}_t, t)}{\lambda} \ell(\mathbf{x}_t) - f(\mathbf{x}_t)^\top \nabla_{\mathbf{x}} \Psi(\mathbf{x}_t, t) \\ &\quad - \frac{\lambda}{2 \Psi(\mathbf{x}_t, t)} \nabla_{\mathbf{x}} \Psi(\mathbf{x}_t, t)^\top \mathbf{G}(\mathbf{x}_t) \mathbf{R}^{-1} \mathbf{G}(\mathbf{x}_t)^\top \nabla_{\mathbf{x}} \Psi(\mathbf{x}_t, t) \\ &\quad + \frac{1}{2 \Psi(\mathbf{x}_t, t)} \nabla_{\mathbf{x}} \Psi(\mathbf{x}_t, t)^\top \mathbf{B}(\mathbf{x}_t) \mathbf{B}(\mathbf{x}_t)^\top \nabla_{\mathbf{x}} \Psi(\mathbf{x}_t, t) \\ &\quad - \frac{1}{2} \text{tr} \left(\mathbf{B}(\mathbf{x}_t) \mathbf{B}(\mathbf{x}_t)^\top \nabla_{\mathbf{x}\mathbf{x}}^2 \Psi(\mathbf{x}_t, t) \right). \end{aligned} \quad (30)$$

Under the noise-control matching assumption

$$\mathbf{B}(\mathbf{x}_t) \mathbf{B}(\mathbf{x}_t)^\top = \lambda \mathbf{G}(\mathbf{x}_t) \mathbf{R}^{-1} \mathbf{G}(\mathbf{x}_t)^\top, \quad (31)$$

the quadratic gradient terms in (30) cancel, and the desirability function $\Psi(\mathbf{x}_t, t)$ satisfies the following linear partial differential equation:

$$\begin{aligned} \partial_t \Psi(\mathbf{x}_t, t) &= \frac{\Psi(\mathbf{x}_t, t)}{\lambda} \ell(\mathbf{x}_t) - f(\mathbf{x}_t)^\top \nabla_{\mathbf{x}} \Psi(\mathbf{x}_t, t) \\ &\quad - \frac{1}{2} \text{tr} \left(\mathbf{B}(\mathbf{x}_t) \mathbf{B}(\mathbf{x}_t)^\top \nabla_{\mathbf{x}\mathbf{x}}^2 \Psi(\mathbf{x}_t, t) \right), \end{aligned} \quad (32)$$

where the terminal condition is given by

$$\Psi(\mathbf{x}_T, T) = \exp \left(-\frac{\phi(\mathbf{x}_T)}{\lambda} \right) \quad (33)$$

Applying the Feynman-Kac theorem [31] to the linear PDE in (32) yields the solution:

$$\begin{aligned} \Psi(\mathbf{x}_{t_0}, t_0) &= \mathbb{E}_{\mathbb{P}} \left[\exp \left(-\frac{1}{\lambda} \int_{t_0}^T \ell(\mathbf{x}_t) dt \right) \Psi(\mathbf{x}_T, T) \right] \\ &= \mathbb{E}_{\mathbb{P}} \left[\exp \left(-\frac{1}{\lambda} S(\tau) \right) \right], \end{aligned} \quad (34)$$

where the trajectory cost is defined as

$$S(\tau) = \phi(\mathbf{x}_T) + \int_{t_0}^T \ell(\mathbf{x}_t) dt, \quad (35)$$

and $\mathbb{P}$ denotes the probability measure induced by the uncontrolled dynamics

$$d\mathbf{x}_t = f(\mathbf{x}_t) dt + \mathbf{B}(\mathbf{x}_t) d\mathbf{w}_t. \quad (36)$$

Differentiating (34) with respect to $\mathbf{x}_{t_0}$ and using the relation between V and Ψ in (28), the optimal control law (25) can be rewritten as the following continuous-time path integral control law:

$$\mathbf{u}^*(\mathbf{x}_{t_0}, t_0) dt = \mathcal{G}(\mathbf{x}_{t_0}) \frac{\mathbb{E}_{\mathbb{P}} \left[\exp \left(-\frac{1}{\lambda} S(\tau) \right) \mathbf{B}_c(\mathbf{x}_{t_0}) d\mathbf{w}_t \right]}{\mathbb{E}_{\mathbb{P}} \left[\exp \left(-\frac{1}{\lambda} S(\tau) \right) \right]}, \quad (37)$$

where $B_c(\mathbf{x}_{t_0})$ appears because, under the partitioned dynamics in (20), the process noise acts directly only on the controlled component. The matrix

$$\mathcal{G}(\mathbf{x}_{t_0}) = R^{-1} G_c(\mathbf{x}_{t_0})^\top \left(G_c(\mathbf{x}_{t_0}) R^{-1} G_c(\mathbf{x}_{t_0})^\top \right)^{-1} \quad (38)$$

transforms the stochastic term in the controlled dynamics into the corresponding control-space expression.

Applying the Euler–Maruyama discretization with timestep Δt , and defining the controlled-state increment as

$$d\mathbf{x}_{t_i}^c \triangleq \mathbf{x}_{t_i+\Delta t}^c - \mathbf{x}_{t_i}^c, \quad (39)$$

we have

$$B_c(\mathbf{x}_{t_0}) d\mathbf{w}_t \approx d\mathbf{x}_{t_0}^c - f_c(\mathbf{x}_{t_0})\Delta t. \quad (40)$$

Therefore, the discrete-time form of (37) becomes

$$\mathbf{u}^*(\mathbf{x}_{t_0}, t_0) = \mathcal{G}(\mathbf{x}_{t_0}) \frac{\mathbb{E}_{\mathbb{P}} \left[\exp \left(-\frac{1}{\lambda} S(\tau) \right) \left(\frac{d\mathbf{x}_{t_0}^c}{\Delta t} - f_c(\mathbf{x}_{t_0}) \right) \right]}{\mathbb{E}_{\mathbb{P}} \left[\exp \left(-\frac{1}{\lambda} S(\tau) \right) \right]}. \quad (41)$$

In discrete time, the horizon is divided into N intervals as

$$N = \frac{T - t_0}{\Delta t}, \quad t_i = t_0 + i\Delta t, \quad (42)$$

and the trajectory cost is approximated as

$$S(\tau) \approx \phi(\mathbf{x}_{t_N}) + \sum_{i=0}^{N-1} \ell(\mathbf{x}_{t_i}) \Delta t, \quad (43)$$

which corresponds to the accumulation of stage costs $\ell(\mathbf{x}_{t_i})$ and the terminal cost $\phi(\mathbf{x}_{t_N})$ along the trajectory. A fundamental difficulty in evaluating (41) is that sampling from the uncontrolled measure $\mathbb{P}$ is highly inefficient. Since the uncontrolled dynamics contain no task-directed input, the probability of generating low-cost trajectories is typically negligible. To address this issue, we introduce a controlled sampling distribution $\mathbb{Q}$ and rewrite (41) using importance sampling:

$$\mathbf{u}^*(\mathbf{x}_{t_0}, t_0) = \mathcal{G}(\mathbf{x}_{t_0}) \frac{\mathbb{E}_{\mathbb{Q}} \left[\exp \left(-\frac{1}{\lambda} S(\tau) \right) \left(\frac{d\mathbf{x}_{t_0}^c}{\Delta t} - f_c(\mathbf{x}_{t_0}) \right) \frac{p(\tau)}{q(\tau)} \right]}{\mathbb{E}_{\mathbb{Q}} \left[\exp \left(-\frac{1}{\lambda} S(\tau) \right) \frac{p(\tau)}{q(\tau)} \right]}. \quad (44)$$

To evaluate the likelihood ratio $p(\tau)/q(\tau)$ in (44), let $p(\tau)$ denote the trajectory probability under the uncontrolled dynamics and $q(\tau)$ the probability under the controlled sampling process with nominal input $\mathbf{u}_{t_i}$. We define the auxiliary variables

$$\mathbf{z}_i := \frac{d\mathbf{x}_{t_i}^c}{\Delta t} - f_c(\mathbf{x}_{t_i}), \quad \boldsymbol{\mu}_i := G_c(\mathbf{x}_{t_i}) \mathbf{u}_{t_i}. \quad (45)$$

Under the Euler–Maruyama discretization of (19) with $\epsilon_{t_i} \sim \mathcal{N}(\mathbf{0}, I)$, the controlled-state increment is

$$d\mathbf{x}_{t_i}^c = f_c(\mathbf{x}_{t_i}) \Delta t + G_c(\mathbf{x}_{t_i}) \mathbf{u}_{t_i} \Delta t + B_c(\mathbf{x}_{t_i}) \epsilon_{t_i} \sqrt{\Delta t}, \quad (46)$$

from which the auxiliary variables satisfy

$$\mathbf{z}_i = G_c \mathbf{u}_{t_i} + \frac{B_c \epsilon_{t_i}}{\sqrt{\Delta t}}, \quad \mathbf{z}_i - \boldsymbol{\mu}_i = \frac{B_c \epsilon_{t_i}}{\sqrt{\Delta t}}. \quad (47)$$

For a discretized trajectory $\tau = \{\mathbf{x}_{t_0}, \dots, \mathbf{x}_{t_N}\}$, the path probability factorizes as $p(\tau) = \prod_i p(\mathbf{x}_{t_{i+1}} | \mathbf{x}_{t_i})$, and analogously for $q(\tau)$. Each one-step transition density is Gaussian with the same covariance Σ_i ; p and q differ only in their means. The respective densities are therefore

$$p(\tau) \propto \exp \left(-\frac{\Delta t}{2} \sum_{i=0}^{N-1} \mathbf{z}_i^\top \Sigma_i^{-1} \mathbf{z}_i \right), \quad (48)$$

$$q(\tau) \propto \exp \left(-\frac{\Delta t}{2} \sum_{i=0}^{N-1} (\mathbf{z}_i - \boldsymbol{\mu}_i)^\top \Sigma_i^{-1} (\mathbf{z}_i - \boldsymbol{\mu}_i) \right),$$

where $\Sigma_i = B_c B_c^\top \Delta t = \lambda G_c R^{-1} G_c^\top \Delta t$ by the matching condition (31). Expanding the quadratic forms and taking the ratio gives

$$\frac{p(\tau)}{q(\tau)} = \exp \left[-\frac{\Delta t}{2} \sum_{i=0}^{N-1} (2\boldsymbol{\mu}_i^\top \Sigma_i^{-1} (\mathbf{z}_i - \boldsymbol{\mu}_i) + \boldsymbol{\mu}_i^\top \Sigma_i^{-1} \boldsymbol{\mu}_i) \right]. \quad (49)$$

We now simplify (49). Assuming that $G_c(\mathbf{x}_{t_i})$ is square and invertible, the inverse covariance is

$$\Sigma_i^{-1} = \frac{1}{\lambda \Delta t} G_c(\mathbf{x}_{t_i})^{-\top} R G_c(\mathbf{x}_{t_i})^{-1}. \quad (50)$$

Substituting (45), (47), and (50) into the quadratic terms of (49), the G_c factors cancel and the two terms become

$$\boldsymbol{\mu}_i^\top \Sigma_i^{-1} (\mathbf{z}_i - \boldsymbol{\mu}_i) = \frac{1}{\lambda \Delta t} \mathbf{u}_{t_i}^\top R G_c^{-1} \frac{B_c \epsilon_{t_i}}{\sqrt{\Delta t}}, \quad (51)$$

$$\boldsymbol{\mu}_i^\top \Sigma_i^{-1} \boldsymbol{\mu}_i = \frac{1}{\lambda \Delta t} \mathbf{u}_{t_i}^\top R \mathbf{u}_{t_i}.$$

The quantity $G_c^{-1} B_c \epsilon_{t_i} / \sqrt{\Delta t}$ in the first line of (51) represents the diffusion noise projected onto the control space. We define the control-space perturbation as follows:

$$\delta \mathbf{u}_{t_i} := G_c(\mathbf{x}_{t_i})^{-1} \frac{B_c(\mathbf{x}_{t_i}) \epsilon_{t_i}}{\sqrt{\Delta t}}, \quad (52)$$

where its covariance under the matching condition (31) is

$$\text{Cov}(\delta \mathbf{u}_{t_i}) = \frac{1}{\Delta t} G_c^{-1} B_c B_c^\top G_c^{-\top} = \frac{\lambda R^{-1}}{\Delta t}. \quad (53)$$

Substituting (51) and (52) into (49), the likelihood ratio simplifies to

$$\frac{p(\tau)}{q(\tau)} = \exp \left[-\frac{1}{\lambda} \sum_{i=0}^{N-1} \left(\mathbf{u}_{t_i}^\top R \delta \mathbf{u}_{t_i} + \frac{1}{2} \mathbf{u}_{t_i}^\top R \mathbf{u}_{t_i} \right) \right]. \quad (54)$$

Absorbing this ratio into the exponential weight in (44) yields

$$\exp\left(-\frac{1}{\lambda}S(\tau)\right) \frac{p(\tau)}{q(\tau)} = \exp\left(-\frac{1}{\lambda}\tilde{S}(\tau)\right), \quad (55)$$

where the augmented trajectory cost is

$$\tilde{S}(\tau) = \phi(\mathbf{x}_{t_N}) + \sum_{i=0}^{N-1} \left[\ell(\mathbf{x}_{t_i}) \Delta t + \mathbf{u}_{t_i}^\top \mathbf{R} \delta \mathbf{u}_{t_i} + \frac{1}{2} \mathbf{u}_{t_i}^\top \mathbf{R} \mathbf{u}_{t_i} \right]. \quad (56)$$

Since $G_c(\mathbf{x}_{t_i})$ is square and invertible, the matrix (38) reduces to $\mathcal{G}(\mathbf{x}_{t_0}) = G_c(\mathbf{x}_{t_0})^{-1}$, and the composite term inside (44) becomes

$$\mathcal{G}(\mathbf{x}_{t_0}) \left(\frac{d\mathbf{x}_{t_0}^c}{\Delta t} - f_c(\mathbf{x}_{t_0}) \right) = G_c(\mathbf{x}_{t_0})^{-1} \mathbf{z}_0 = \mathbf{u}_{t_0} + \delta \mathbf{u}_{t_0}. \quad (57)$$

Substituting (55) and (57) into (44), the term $\frac{1}{2} \sum_i \mathbf{u}_{t_i}^\top \mathbf{R} \mathbf{u}_{t_i}$ in (56) does not depend on τ and appears as a common multiplicative factor $\exp\left(-\frac{1}{2\lambda} \sum_i \mathbf{u}_{t_i}^\top \mathbf{R} \mathbf{u}_{t_i}\right)$ in both the numerator and denominator, which cancels upon division. Defining the reduced augmented cost

$$\hat{S}(\tau) = \phi(\mathbf{x}_{t_N}) + \sum_{i=0}^{N-1} \left[\ell(\mathbf{x}_{t_i}) \Delta t + \mathbf{u}_{t_i}^\top \mathbf{R} \delta \mathbf{u}_{t_i} \right], \quad (58)$$

the optimal control law becomes

$$\mathbf{u}_{t_0}^* = \mathbf{u}_{t_0} + \frac{\mathbb{E}_{\mathbb{Q}} \left[\exp\left(-\frac{1}{\lambda} \hat{S}(\tau)\right) \delta \mathbf{u}_{t_0} \right]}{\mathbb{E}_{\mathbb{Q}} \left[\exp\left(-\frac{1}{\lambda} \hat{S}(\tau)\right) \right]}. \quad (59)$$

Approximating the expectations with K sampled trajectories yields the MC MPPI update:

$$\mathbf{u}_{t_0}^* \approx \mathbf{u}_{t_0} + \sum_{k=1}^K w^{(k)} \delta \mathbf{u}_{t_0}^{(k)}, \quad (60)$$

with normalized importance weights

$$w^{(k)} = \frac{\exp\left(-\frac{1}{\lambda} \hat{S}(\tau^{(k)})\right)}{\sum_{j=1}^K \exp\left(-\frac{1}{\lambda} \hat{S}(\tau^{(j)})\right)}. \quad (61)$$

Equations (60)–(61) complete the path integral derivation of the MPPI update law. A detailed comparison with the information-theoretic derivation in Section 2.1, including the differing assumptions and their practical consequences, is provided in Section 2.3.

2.3. Comparative Analysis of the Two Derivations

Sections 2.1 and 2.2 derive the MPPI update law from two fundamentally different mathematical frameworks, yet

both produce the same importance-weighted control update for each time step t_i in the planning horizon

$$\mathbf{u}_{t_i}^* \approx \hat{\mathbf{u}}_{t_i} + \sum_{k=1}^K w^{(k)} \delta \mathbf{u}_{t_i}^{(k)}, \quad i = 0, \dots, N-1, \quad (62)$$

where $w^{(k)}$ are normalized exponential weights over an augmented trajectory cost. This mathematical equivalence is analyzed in Section 2.3.1. Section 2.3.2 examines the differing structural assumptions, and Section 2.3.3 relates MPPI to the open-loop, closed-loop, and deterministic formulations of optimal control.

2.3.1. Mathematical Correspondence Between the Two Derivations

The equivalence of the two MPPI derivations rests on three interlocking correspondences between the objective functions, the sampling distributions, and the augmented cost terms. Throughout this analysis we adopt the normalization $\Delta t = 1$ following [27], which places the information-theoretic and path integral formulations on a common discrete-time footing.

Equivalence of the objective functions. Both derivations arrive at the same mathematical expression for the optimal value. The information-theoretic formulation defines the free energy (5) as $\mathcal{F}_\lambda = -\lambda \log \mathbb{E}_{\mathbb{P}}[\exp(-S/\lambda)]$ and shows that the SOC objective attains this value at optimality. The path integral formulation combines the Feynman–Kac solution (34) with the exponential transformation (28) to obtain

$$V(\mathbf{x}_{t_0}, t_0) = -\lambda \log \mathbb{E}_{\mathbb{P}} \left[\exp\left(-\frac{S(\tau)}{\lambda}\right) \right], \quad (63)$$

which has the same functional form as $\mathcal{F}_\lambda$. This identity holds for control-affine systems satisfying the matching condition (31), as originally shown by Kappen [28, 29]. For the two formulations to produce the same numerical value, however, the base measures $\mathbb{P}$ must induce the same state-trajectory distribution. This requirement is fulfilled through the correspondence of the sampling distributions, established next.

Correspondence of the sampling distributions. In the path integral formulation, the control-space perturbation defined in (52), which requires the invertibility of G_c , has covariance (53) that under $\Delta t = 1$ reduces to

$$\text{Cov}(\delta \mathbf{u}^{\text{PI}}) = \lambda \mathbf{R}^{-1}. \quad (64)$$

In the information-theoretic formulation, $\delta \mathbf{u}^{\text{IT}} \sim \mathcal{N}(\mathbf{0}, \Sigma)$. Selecting

$$\Sigma = \lambda \mathbf{R}^{-1} \quad \Longleftrightarrow \quad \mathbf{R} = \lambda \Sigma^{-1} \quad (65)$$

makes the two perturbation distributions identical. Under this identification, the base measures $\mathbb{P}$ in the information-theoretic and path integral formulations generate the same state-trajectory distribution, and the value function identity (63) guarantees that both formulations yield the same optimal control.

Equivalence of the augmented cost terms. The cross-terms in the two importance weights coincide under (65). The information-theoretic weight (17) involves $\mathbf{v}_t$ rather than $\delta\mathbf{u}_t$ directly. Substituting $\mathbf{v}_t = \hat{\mathbf{u}}_t + \delta\mathbf{u}_t$ with base distribution $\tilde{\mathbf{u}} = \mathbf{0}$

$$\lambda \hat{\mathbf{u}}_t^\top \Sigma^{-1} \mathbf{v}_t = \lambda \hat{\mathbf{u}}_t^\top \Sigma^{-1} \hat{\mathbf{u}}_t + \lambda \hat{\mathbf{u}}_t^\top \Sigma^{-1} \delta\mathbf{u}_t. \quad (66)$$

The quadratic term $\lambda \hat{\mathbf{u}}_t^\top \Sigma^{-1} \hat{\mathbf{u}}_t$ does not depend on the sampled trajectory and cancels between numerator and denominator upon normalization. The remaining cross-term becomes, via (65),

$$\lambda \hat{\mathbf{u}}_t^\top \Sigma^{-1} \delta\mathbf{u}_t = \hat{\mathbf{u}}_t^\top \mathbf{R} \delta\mathbf{u}_t, \quad (67)$$

which matches the cross-term $\mathbf{u}_t^\top \mathbf{R} \delta\mathbf{u}_t$ in the path integral reduced augmented cost (58). The state-dependent costs also agree, since under $\Delta t = 1$ the information-theoretic cost $S(\mathbf{V}) = \phi(\mathbf{x}_T) + \sum_t \ell(\mathbf{x}_t)$ and the path integral cost $\phi(\mathbf{x}_{t_N}) + \sum_i \ell(\mathbf{x}_{t_i})$ evaluate to the same numerical value for corresponding trajectories.

In summary, under the matching condition (31), the identification (65), and the invertibility of $G_c(\mathbf{x}_t)$, the two derivations share the same optimal value (63), the same perturbation distribution, and identical augmented costs, establishing their mathematical equivalence.

2.3.2. Assumptions Underlying the Two Derivations

Although the two derivations yield the same update law under the conditions identified in Section 2.3.1, they impose fundamentally different structural requirements on the system. The path integral derivation requires control-affine dynamics (19) and the noise–control matching condition (31). Without the matching condition, the quadratic gradient terms in the nonlinear HJB equation (26) do not cancel, preventing the linearization that enables the path integral representation of the value function.

Concretely, the matching condition (31) imposes two requirements, first identified by Kappen [28, 29]. First, stochastic disturbances may enter the dynamics only through states that also possess control authority. In the partitioned dynamics (20), this is encoded by restricting both G and B to act exclusively on the controlled component $\mathbf{x}_t^c$, while the passive states $\mathbf{x}_t^p$ remain free of both control input and diffusion. This is a genuine structural restriction that cannot be resolved through parameter tuning. Second, within the controlled subspace, the control cost matrix R must be scaled to match the noise covariance through $B_c B_c^\top = \lambda G_c R^{-1} G_c^\top$, creating an inverse relationship between noise magnitude and control cost. Directions with large noise variance must have correspondingly cheap control authority, while directions with small noise require expensive control. This second requirement is less restrictive in practice, because R is a design parameter that can be chosen to match the noise profile of the system.

The path integral control law in its general form (37) does not require $G_c(\mathbf{x}_t)$ to be square or invertible, as the matrix G (38) accommodates non-square G_c without requiring its inversion [30]. However, the simplified derivation in

Section 2.2 assumes that G_c is square and invertible, which is needed to define the control-space perturbation (52) and to reduce G to G_c^{-1} . This assumption is also required for the sampling distribution correspondence and the augmented cost equivalence established in Section 2.3.1. Requiring G_c to be square means that the number of control inputs m must equal the dimension of the controlled states $n - \ell$, which excludes overactuated systems ($m > n - \ell$) where redundant actuators provide more control authority than the controlled-state dimension, and underactuated systems ($m < n - \ell$) where certain controlled-state directions lack direct actuation.

The information-theoretic derivation, by contrast, requires only that the dynamics $f(\mathbf{x}_t, \mathbf{v}_t)$ in (1) be measurable and that $S(\mathbf{V})$ in (3) be finite. No structural constraint on the form of the dynamics is imposed, and no restriction on the relationship between the number of control inputs and state dimensions applies. This substantially broader applicability is the primary reason why modern MPPI implementations adopt the information-theoretic framework [27].

2.3.3. Open-Loop, Closed-Loop, and Deterministic Perspectives

A critical distinction concerns whether the underlying SOC problem is formulated in an open-loop or closed-loop sense. Homburger *et al.* [32] recently provided a rigorous treatment of this issue, identifying three problem classes and analyzing how MPPI relates to each.

Open-Loop Stochastic Optimal Control (OL-SOC). In the open-loop formulation, the entire control sequence $\mathbf{U} = (\mathbf{u}_0, \dots, \mathbf{u}_{T-1})$ is determined at the initial time before any disturbance is realized, and the controller does not adapt to intermediate states. The information-theoretic derivation in Section 2.1 corresponds naturally to this setting, as it optimizes over $\mathbf{U}$ by minimizing the KL divergence between the optimal and controlled trajectory distributions. A practical limitation arises because, for nonlinear dynamics, the optimal distribution $\mathbb{Q}^*$ in (12) is generally non-Gaussian and cannot be represented exactly by the Gaussian proposal $\mathbb{Q}$.

Closed-Loop Stochastic Optimal Control (CL-SOC). In the closed-loop formulation, each control input $\mathbf{u}_t$ is selected as a function of the realized state $\mathbf{x}_t$, corresponding to optimization over state-feedback policies. The path integral derivation in Section 2.2 addresses this setting by solving for the value function $V(\mathbf{x}, t)$, from which the state-feedback law $\mathbf{u}^* = -R^{-1} G^\top \nabla V$ is obtained. For control-affine systems satisfying the matching condition (31), the path integral formulation provides an exact representation of the CL-SOC value function. The closed-loop formulation also gives rise to phenomena absent in the open-loop setting. In environments where multiple trajectory modes of equal cost exist, the optimal closed-loop policy can defer the commitment to a specific mode, since the policy will select the appropriate direction once future noise resolves the ambiguity [28]. In contrast, the open-loop formulation must commit to a single

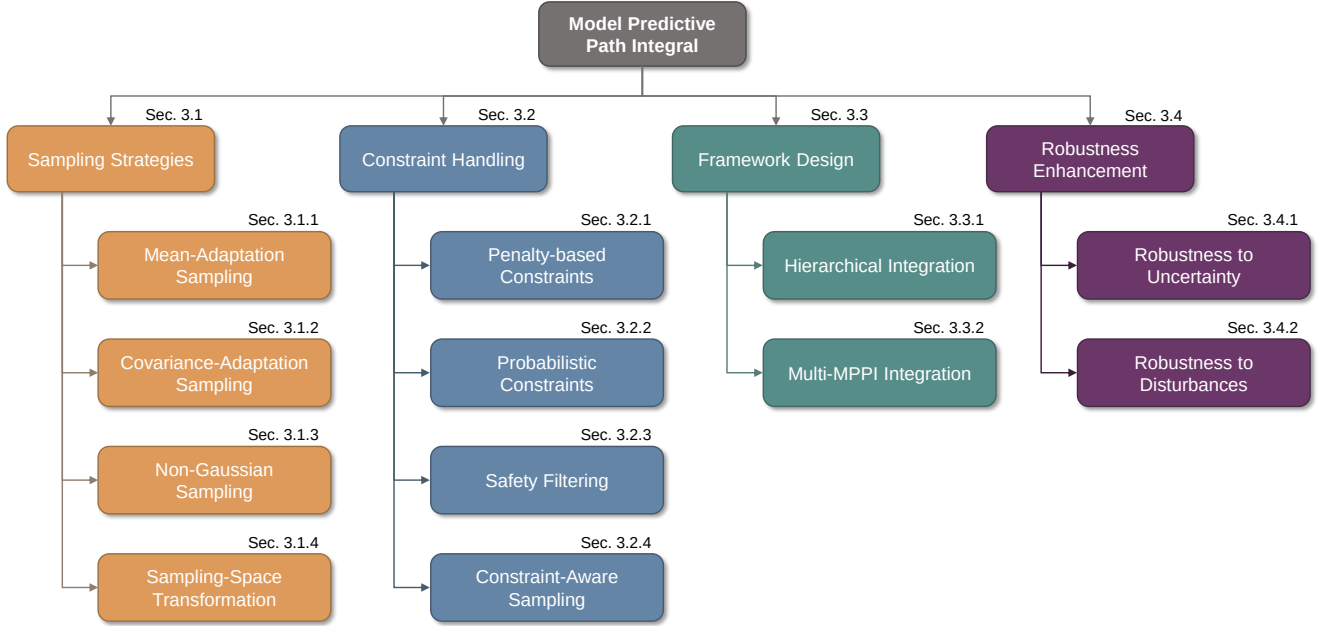


Figure 2: Taxonomy of methodological themes in MPPI.

control sequence at the initial time, which can produce a compromise control that points between the modes.

Deterministic Setting and Receding-Horizon Execution. Because of its broader applicability to general nonlinear systems, modern MPPI is commonly implemented through the information-theoretic formulation. In this setting, the stochastic perturbations are typically not a model of environmental uncertainty but rather an exploration mechanism injected around a nominal control sequence for trajectory optimization. From this perspective, MPPI can be viewed as a sampling-based approximate solver for the underlying deterministic problem, where the injected noise serves purely as an exploration device. For continuously differentiable, unconstrained problems, Homburger *et al.* [32] showed that this discrepancy is controlled by the sampling noise level. If α scales the standard deviation of the sampling distribution, then the control error decreases as $O(\alpha^2)$ and the value function error as $O(\alpha^4)$.

In practice, MPPI has been widely and successfully applied to systems where the matching condition (31) does not hold exactly. This robustness can be attributed to the receding-horizon execution: at each cycle, MPPI computes an open-loop control sequence, applies only the first element, updates the state from the latest measurement, and resolves the optimization over the shifted horizon. For control-affine systems satisfying the matching condition, the first control of the OL-SOC and CL-SOC solutions coincide [32], so each replanning cycle recovers the closed-loop optimal action at the current state. Even when this condition is not met, repeated replanning induces closed-loop behavior by correcting approximation errors at every control cycle. However, this empirical error-correction mechanism should not be conflated with a formal closed-loop stability guarantee.

Establishing such guarantees for MPPI remains an open problem and is discussed further in Section 5.

3. Methodological Themes of MPPI

This section provides a structured review of methodological advances in MPPI research. To bring clarity to the fragmented landscape of MPPI variants, we categorize existing approaches into four key themes: **(i) Sampling Strategies**, **(ii) Constraint Handling**, **(iii) Framework Design**, and **(iv) Robustness Enhancement**. As illustrated in Fig. 2, these four overarching themes encompass the principal methodological directions along which approaches built upon MPPI have been developed and diversified. Within each theme, existing works can be further subdivided into more specific categories that capture distinct algorithmic trends. For example, Sampling Strategies in this survey are categorized into mean-adaptation sampling, covariance-adaptation sampling, non-Gaussian sampling, and sampling-space transformation. The following subsections elaborate on each theme and its representative methods in detail.

3.1. Sampling Strategies

One of the core factors determining MPPI performance is the quality of sampled trajectories. Since MPPI relies on importance-weighted averaging over sampled rollouts, its efficiency depends on whether low-cost trajectories are sufficiently represented within a limited sample budget. This reliance is explicitly reflected in the MPPI update law (18), where the nominal control sequence is refined by aggregating perturbations weighted by their exponentiated costs (17). However, as discussed in Section 2.3, the optimal distribution $\mathbb{Q}^*$ in (12) is generally non-Gaussian for nonlinear systems, and a naive Gaussian proposal cannot approximate

Table 1

Qualitative comparison of sampling strategies in MPPI across five performance criteria.

Approach direction	Representative methods	Sample efficiency	Computational efficiency	Multi-modal exploration	Trajectory smoothness	Domain generality
Planner-guided mean adaptation (Sec. 3.1.1)	RRT / A*-guided [33, 34]	○	×	×	×	×
Learning-based mean adaptation (Sec. 3.1.1)	Diffusion / CVAE [35, 36]	○	△	○	×	△
	RL policy [37, 38, 39]	○	○	×	×	△
Gradient-based sample transport (Sec. 3.1.1)	SVGD-based [40]	○	×	×	×	○
Temporally-correlated noise design (Sec. 3.1.2)	Temporal correlation / Colored noise sampling [41, 42]	△	○	×	○	○
Online exploration adaptation (Sec. 3.1.2)	Error-adaptive scaling [43]	△	○	×	△	○
	Annealing / AIS [44, 45, 46]	○	×	×	△	○
Non-Gaussian perturbation (Sec. 3.1.3)	Log-normal / Uniform [47, 48]	△	○	△	×	○
	Configuration-uniform [49, 50]	△	△	○	×	△
Task-aware proposal design (Sec. 3.1.3)	Multi-policy proposal / Normalizing flow [51, 52, 53, 54]	○	△	○	×	△
Sampling-space transformation (Sec. 3.1.4)	Output-space / Tangent-space [55, 56]	○	△	×	×	△
	Adaptive space switching [57]	○	○	×	×	△
	Derivative-action lifting [58]	△	○	×	○	○

○: High, △: Medium, ×: Low.

it closely. In practice, this leads to limited coverage of promising regions, poor trajectory quality, and degraded performance in complex tasks. To address these limitations, numerous studies have developed advanced sampling strategies that improve sample efficiency, trajectory quality, or both.

As summarized in Table 1, these strategies exhibit systematic trade-offs across the five evaluation criteria. Incorporating task-specific knowledge through external planners, learned priors, or guidance policies generally improves sample efficiency by concentrating rollouts in promising regions. However, planner-guided approaches and gradient-based sample transport methods incur significant computational overhead due to the cost of running auxiliary planners or iterative particle updates at each control cycle, and adaptive importance sampling and annealing strategies similarly require multiple sequential refinement passes that increase per-iteration cost. Learning-based approaches that rely on offline-trained models, such as diffusion or Conditional Variational Autoencoder (CVAE) priors, amortize most computation into the training phase and thus maintain moderate online cost, though they introduce dependencies on training data and environment-specific models that can limit domain generality. Methods that reshape the sampling distribution without task-specific priors, such as temporally-correlated noise design and error-adaptive scaling, preserve broad applicability and low computational cost, though their undirected coverage places a smaller fraction of samples in informative regions.

Trajectory smoothness and multi-modal exploration constitute two additional axes along which the strategies diverge. Only methods that explicitly restructure the temporal noise correlation or lift the sampling domain to derivative-action coordinates embed smoothness directly into the sampling process, whereas mean-adaptation and non-Gaussian

methods do not alter the temporal structure of the perturbation noise and are therefore not designed to address smoothness. The degree to which a method achieves multi-modal exploration depends on how diversity is introduced, ranging from broadening the noise distribution through heavy-tailed or uniform sampling to employing generative priors, configuration-space uniform sampling, or multi-policy proposals that can explicitly represent distinct solution modes.

3.1.1. Mean-Adaptation Sampling

A major limitation of the standard MPPI control framework lies in the use of a naive Gaussian sampling distribution centered around the control sequence obtained in the previous iteration. Such uninformed sampling typically results in inefficient trajectory exploration, particularly when applied to complex environments or high-dimensional nonlinear dynamical systems. To mitigate this limitation, a class of mean-adaptation sampling methods has been developed to directly modify the nominal control sequence $\mathbf{u}_t$ in the MPPI update (18), thereby biasing the proposal distribution towards dynamically feasible and cost-efficient regions. By explicitly defining or adaptively updating a reference mean control sequence, these methods shift the center of the sampling distribution to guide exploration towards more promising trajectories.

A straightforward way to implement this approach is to inject an external path planner that provides a nominal control sequence serving as the mean of the sampling distribution. For example, Tao *et al.* [33] proposed a Rapidly-exploring Random Tree (RRT)-guided MPPI algorithm, in which a nominal trajectory computed by the RRT is converted into a corresponding control sequence and subsequently used as the mean of the Gaussian proposal distribution. Likewise, Yang *et al.* [34] extended the same concept through a traversability-aware Hybrid A* planner operating on a 2.5-dimensional cost map, which generates

a nominal control sequence that improves vehicle stability and reduces pitch and roll oscillations on uneven terrains. Although the integration of deterministic motion planners provides initial guidance and accelerates early convergence, the planner-guided MPPI approach inherently requires an explicit environmental model.

This model dependence led to the development of data-driven mean generation techniques, where the sampling mean is inferred from data rather than constructed analytically. Huang *et al.* [35] introduced Diffusion-MPPI, which employs a conditional diffusion model to generate a nominal trajectory that initializes the mean of the sampling distribution from environmental point-cloud observations. The diffusion model provides robust distributional guidance but requires iterative denoising, which increases computational cost during online control. In contrast, Kusumoto *et al.* [36] proposed Informed Information-Theoretic MPC (IIT-MPC), which adopts a single-shot latent decoding approach based on a CVAE to generate a context-aware mean trajectory with significantly reduced online computational overhead.

Building on the concept of learning-based guidance, RL has also been integrated into MPPI to improve the quality of trajectory sampling. For instance, Zhao *et al.* [37] proposed an RL-MPPI framework in which an RL agent predicts an action that serves as a guide for the MPPI sampling distribution. Similarly, Qu *et al.* [38] presented RL-driven MPPI, where an offline-trained RL policy is used to initialize the control distribution and generate informative rollouts for online refinement. While these approaches improve sample efficiency and rollout quality by leveraging learned priors, their behavior remains strongly influenced by the pretrained policy. To address online adaptation more explicitly, Wang *et al.* [39] proposed Residual-MPPI, which not only uses a prior policy for initialization but also incorporates the policy's log-likelihood into the MPPI objective, thereby enabling online policy customization that balances preservation of prior behaviors with adaptation to new task objectives.

In contrast to the above data-driven approaches that preserve sampling diversity through generative priors or learned policies, deterministic mode-seeking methods have been explored to directly concentrate samples around dominant low-cost solutions. Honda *et al.* [40] proposed Stein Variational Guided MPPI (SVG-MPPI), which iteratively transports samples towards low-cost regions using Stein Variational Gradient Descent (SVGD). Unlike diffusion- or CVAE-based methods that preserve sample diversity, SVG-MPPI performs deterministic mode-seeking transport, progressively aligning the sampling mean with a dominant feasible mode. This improves convergence stability in scenarios with sharply separated solution modes.

3.1.2. Covariance-Adaptation Sampling

Regulating the covariance structure of the sampling distribution is a key strategy that goes beyond merely reducing noise, aiming to improve the quality of exploration and

convergence stability. These approaches can be broadly categorized into methods that design the covariance structure to introduce temporal correlation, and those that dynamically regulate the magnitude of variance to balance exploration and exploitation.

One class of approaches focuses on introducing temporal coherence to the sampled rollouts. Time-Correlated MPPI (TC-MPPI), introduced by Lee *et al.* [41], incorporates a quadratic cost on action derivatives into the optimal control problem. This cost term is mathematically absorbed into the covariance matrix, creating a non-diagonal inverse covariance matrix that encodes correlations between temporally adjacent control inputs. As a result, the sampling process itself embeds temporal correlation, leading to dynamically feasible and smoother trajectories. Low-Frequency MPPI, proposed by Vlahov *et al.* [42], achieves this goal differently by sampling colored noise in the frequency domain. In this approach, the variance of each frequency component is scaled according to a power law to intentionally suppress high-frequency components. The resulting control samples, generated via an inverse Fast Fourier Transform (iFFT), are inherently smooth and dominated by low frequencies, providing trajectories that are better suited for long-horizon exploration.

Another class of approaches regulates the magnitude of the variance based on real-time feedback or a predefined schedule. Dynamic-Variance MPPI (DV-MPPI), presented by Kim *et al.* [43], employs an intuitive mechanism that scales the variance in proportion to the real-time tracking error using a hyperbolic tangent function. This mechanism allows for aggressive exploration when the error is large, and stable fine-grained control when the error is small. However, because variance adaptation is triggered only after tracking error is observed, DV-MPPI reacts to disturbances rather than anticipating them, which may temporarily limit exploration in rapidly changing environments. As an alternative variance modulation strategy, Xue *et al.* [44] proposed Diffusion-Inspired Annealing for Legged MPC (DIAL-MPC), which avoids reliance on feedback by employing a proactive annealing schedule. Rather than waiting for error feedback, DIAL-MPC progressively decreases the variance through a diffusion-inspired dual-loop process within a single control step, maintaining exploratory capability while guiding rollouts towards local convergence.

Beyond these two covariance-level adaptation strategies, a distinct line of work seeks to optimize the entire sampling distribution directly. Model Predictive Optimized Path Integral (MPOPI), proposed by Asmar *et al.* [45], formulates the control sequence as a single joint vector and applies Adaptive Importance Sampling (AIS). This process iteratively updates both the mean and covariance of the joint distribution within a single control step, significantly improving sample efficiency by more effectively targeting high-reward regions. Building upon this framework, Keshavarz *et al.* [46] extended MPOPI to whole-body locomotion control of legged robots by combining CEM and CMA within the

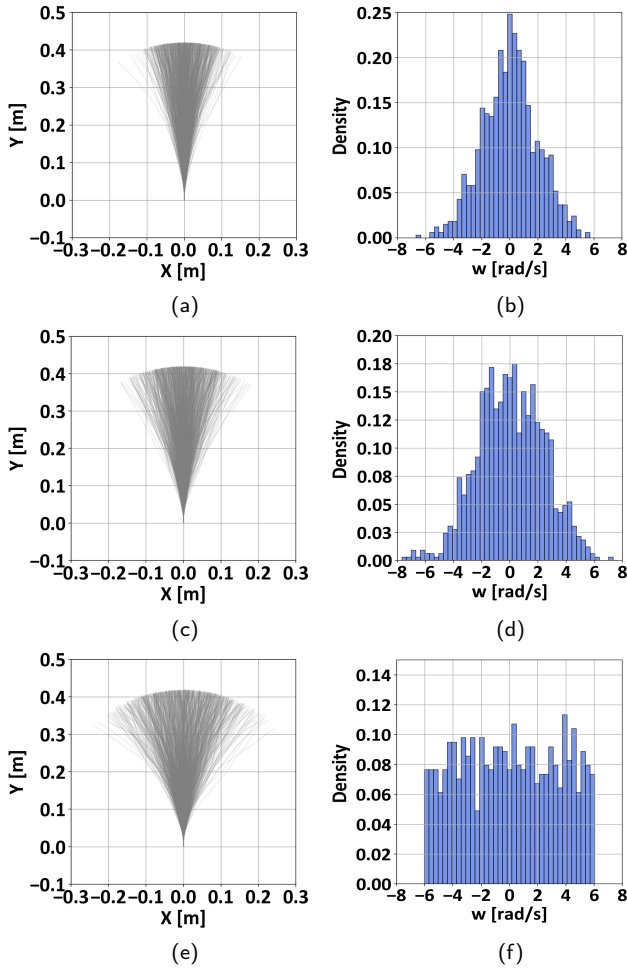


Figure 3: Visualization of 1,000 sampled control noises and the corresponding trajectories under different sampling distributions. (a) Histogram of control-input noise sampled from a Gaussian distribution based on [27]. (b) Resulting trajectories obtained from the Gaussian sampling, where the control noise follows $\delta u \sim \mathcal{N}(0, 2.0^2)$. (c) Histogram of control-input noise sampled from an NLN distribution based on [47]. (d) Resulting trajectories obtained from the NLN sampling, which consists of a normal component $\delta u^n \sim \mathcal{N}(0, 2.0^2)$ and a lognormal component $\delta u^{ln} \sim \text{Log}\mathcal{N}(0.2, 0.1^2)$. (e) Histogram of control-input noise sampled from a random uniform distribution inspired by [48]. (f) Resulting trajectories obtained from the uniform sampling with a range of $[-6, 6]$.

iterative refinement process, demonstrating improved sample efficiency in high-dimensional joint-space exploration for quadrupedal locomotion and loco-manipulation tasks.

3.1.3. Non-Gaussian Sampling

Section 2 provides an information-theoretic interpretation of the stochastic optimal control problem and introduces the free-energy inequality shown in (8). When both $\mathbb{P}$ and $\mathbb{Q}$ are assumed to be Gaussian, the problem becomes equivalent to the SOC formulation; however, this assumption is not essential. From an information-theoretic standpoint, the optimal distribution $\mathbb{Q}^*$ that minimizes the free-energy bound remains invariant regardless of the parametric form of $\mathbb{P}$ and

$\mathbb{Q}$. Even when $\mathbb{P}$ and $\mathbb{Q}$ are not Gaussian, the KL minimization problem that seeks to align the controlled distribution $\mathbb{Q}$ with the optimal distribution $\mathbb{Q}^*$ remains equivalent under the same information-theoretic formulation. By extending the sampling distribution $\mathbb{Q}$ beyond the Gaussian family to a non-Gaussian form, the search space of the controller is generalized to a broader probabilistic control space, thereby enhancing its capability to approximate optimality across diverse distributional structures.

Early work focused on replacing Gaussian noise δu_t in the MPPI update (18) with analytically designed non-Gaussian perturbations to increase sampling diversity. Mohamed *et al.* [47] proposed Log-MPPI, in which the control perturbation δu_t is constructed from a Normal–LogNormal (NLN) mixture distribution to introduce skewness and heavy tails:

$$\delta u_t = \delta u_t^n \cdot \delta u_t^{ln}, \quad (68)$$

where

$$\delta u_t^n \sim \mathcal{N}(\mathbf{0}, \Sigma^n), \quad \delta u_t^{ln} \sim \text{Log}\mathcal{N}(\mu^{ln}, \Sigma^{ln}). \quad (69)$$

This mixture-based sampling expands the exploration range beyond Gaussian perturbations and improves robustness against local minima. Extending this direction, Seo *et al.* [48] replaced Gaussian noise with a uniform distribution-based deterministic sampling scheme, significantly reducing the number of samples required for real-time CoM trajectory generation.

Fig. 3 illustrates the control-input noise distributions and the resulting state trajectories for Gaussian, NLN, and Uniform sampling schemes using comparable scales. Distinct from the deterministic scheme employed in [48], we adopt random uniform sampling to ensure a fair evaluation under stochastic variability. In terms of trajectory generation, the NLN distribution yields a wider spread due to its skewness, despite sharing the same mean and variance as the Gaussian case. In contrast, the Uniform sampling provides perturbations with uniform density, leading to improved exploration efficiency but still resulting in limited state-space coverage after dynamics propagation. To address this state-space coverage limitation and the tendency of samples to collapse around the nominal action sequence, Poyrazoglu *et al.* [49] proposed C-Uniform trajectory sampling, which explicitly enforces uniform coverage in the configuration space by constructing trajectories that are uniformly distributed over reachable sets. This approach mitigates sample collapse and improves global exploration in motion planning tasks.

However, this method relies on explicit discretization of the configuration space and solves a network-flow optimization at each level-set transition, incurring computational costs that scale cubically with the number of discretized cells. To overcome this limitation, the same authors [50] proposed C-Uniform MPPI (CU-MPPI), which replaces the grid-based computation with a neural network trained via entropy maximization to directly map states to control input probabilities, enabling C-Uniform trajectory generation over

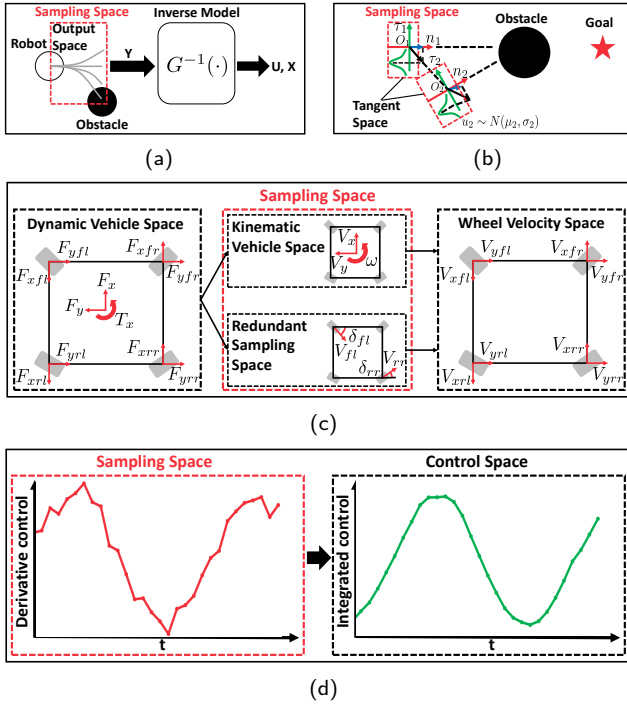


Figure 4: Overview of MPPI approaches that modify or transform the sampling space to improve performance under different objectives. (a) o-MPPI [55] samples directly in the output trajectory space to increase the likelihood of generating samples that satisfy output constraints. (b) MPPI-Tan [56] restricts sampling onto the obstacle-tangent manifold, suppressing normal-direction perturbations that typically lead to collisions in cluttered scenes. (c) MPPI-H [57] switches between low-dimensional kinematic and redundant wheel-space sampling representations to balance sample efficiency and navigation stability for high-DoF swerve-drive robots. (d) SMPPI [58] lifts the sampling space to derivative-action coordinates, producing smooth control sequences without external filtering in rapidly changing environments.

longer horizons without explicit discretization. CU-MPPI leverages these uniformly sampled trajectories to select a low-cost nominal trajectory, which is then used as the sampling mean for subsequent MPPI optimization. Nevertheless, these non-Gaussian and uniform-based sampling schemes still do not incorporate explicit task-aware or safety-aware directional guidance in the state space.

To further improve sampling efficiency, another line of research introduces task-aware bias into the proposal distribution. Homburger *et al.* [51] presented Feature-Based MPPI, which augments the standard Gaussian proposal with auxiliary Gaussian proposals centered on task-specific feature policies such as emergency braking. While each proposal remains Gaussian, sampling from multiple proposals with distinct means achieves an effect analogous to multi-modal exploration. Similarly, Trevisan and Alonso-Mora [52] proposed Biased-MPPI, which generalizes MPPI to arbitrary proposal distributions by fusing multiple ancillary controllers (e.g., goal-seeking, braking, or safety controllers) into the sampling process through importance

weighting, yielding greater robustness under local minima and dynamic disturbances. These methods, however, rely on predefined guidance policies, which may limit adaptability to new environments.

More recent approaches aim to adapt the sampling distribution automatically using data-driven refinement. Flow-based methods such as those by Power and Berenson [53] and Sacks and Boots [54] employ normalizing flows to learn expressive multimodal sampling distributions conditioned on the environment and task context. The approach in [53] amortizes a conditional control posterior and introduces an environment-projection mechanism to address out-of-distribution scenarios. Extending this flow-based conditional sampling framework, the method in [54] performs the online refinement step in a learned latent space, while training the latent sampler so that these updates produce lower-cost control samples.

3.1.4. Sampling-Space Transformation

In contrast to approaches that modify the sampling distribution within the original control-input space, a fundamentally different strategy in MPPI is to transform the sampling space itself. By redefining the representation in which control perturbations are generated, these methods enable exploration to be conducted in a coordinate system that more directly reflects task structure, system geometry, or actuation dynamics.

Yan and Devasia [55] proposed Output-Sampled MPPI (o-MPPI), which replaces input-space sampling with sampling directly in the output space, where task-relevant motions such as lane following or path tracking are more naturally represented. By generating rollouts in this transformed domain and mapping them back through an inverse model, o-MPPI improves the compatibility between sampled rollouts and high-level task objectives, without explicitly enforcing constraints. A subsequent follow-up by the same authors established the theoretical optimality of o-MPPI under an inverse-model-based formulation, showing that output-space sampling can preserve the optimality properties of standard MPPI when cast within an augmented system [59]. For motion planning around nonconvex obstacles, Zhao *et al.* [56] developed MPPI-Tan, which decomposes the control velocity into nominal and tangential components and represents the tangential component in a tangent-space basis defined by the obstacle geometry. This formulation enables explicit optimization of tangential motion across changing local tangent spaces, helping the planner avoid local minima and remain effective in cluttered environments.

To address sampling inefficiency in high-dimensional systems, Aoki *et al.* [57] proposed a switching sampling-space strategy for MPPI that alternates between a reduced kinematic space and a slightly redundant control space. By selecting the sampling representation online according to the tracking condition, the method balances efficiency and stability in high-dimensional vehicle navigation. Beyond switching between sampling spaces, Kim *et al.* [58] proposed Smooth MPPI (SMPPI), which lifts the sampling

Table 2

Qualitative comparison of constraint-handling strategies in MPPI.

Criterion	Penalty-based Constraints (Section 3.2.1)	Probabilistic Constraints (Section 3.2.2)	Safety Filtering (Section 3.2.3)	Constraint-Aware Sampling (Section 3.2.4)
Sample efficiency	$\triangle$	$\triangle$	$\triangle$	$\circ$
Safety guarantees	$\triangle$	$\circ$	$\circ$	$\circ$
Handling non-differentiable costs	$\circ$	$\triangle$	$\times$	$\times$
Handling of uncertainty	$\times$	$\circ$	$\triangle$	$\times$

 $\circ$: High, $\triangle$: Medium, $\times$: Low.

domain to the derivative action space to mitigate chattering. Rather than sampling control inputs directly, the method samples derivative actions and reconstructs the action sequence through temporal integration, while also introducing an action-variation cost to discourage rapid changes along the rollout. As a result, SMPPI generates smoother commands without relying on external smoothing filters. These sampling-space transformation strategies are conceptually summarized in Fig. 4.

3.2. Constraint Handling

MPPI offers the flexibility to encode diverse objectives such as goal tracking, control smoothness, actuation limits, workspace boundaries, and collision avoidance directly through cost terms without requiring model linearization or differentiability. However, since these requirements are typically imposed as soft penalties rather than explicit constraints, MPPI does not provide a formal guarantee of constraint satisfaction, and safety or feasibility violations may still occur. To address this limitation, a wide spectrum of constraint-handling strategies has been developed, ranging from penalty-based formulations to constraint-aware sampling.

As summarized in Table 2, these strategies present distinct trade-offs between constraint flexibility and the strength of safety enforcement. Penalty-based methods offer the broadest applicability by accommodating non-differentiable and black-box cost formulations, but provide only soft safety guarantees that cannot prevent occasional violations. As constraint enforcement becomes more rigorous through probabilistic formulations, safety filtering, or post-sampling correction, stronger safety properties are achieved, though at the cost of requiring differentiable system models or analytically tractable constraint structures. Uncertainty handling constitutes a separate consideration. Only probabilistic methods explicitly reason about stochastic variability at the distributional level, while safety filtering addresses it partially through stochastic barrier formulations. Regarding sample efficiency, constraint-aware sampling directly corrects infeasible samples into the feasible region, thereby increasing the effective fraction of useful samples and achieving higher sample efficiency than the other three categories. The following sections review these approaches in detail.

3.2.1. Penalty-based Constraints

The most intuitive strategy for handling constraints in MPPI is to embed them directly into the cost function as penalty terms. While this soft-constraint approach does not guarantee strict feasibility, it biases the stochastic optimization process towards safer trajectories by assigning higher costs to constraint-violating regions. Penalty-based formulations are particularly attractive in MPPI for several reasons. They impose no differentiability requirements, readily accommodate black-box or neural-network-based costs, and allow heterogeneous and highly expressive constraint objectives to be incorporated in a unified manner. This general strategy has been explored across a diverse range of constraint types, from spatial constraints to more abstract formulations that shape long-horizon behavior.

Among these applications, the penalty-based approach has been extensively applied to manage spatial constraints, particularly for collision avoidance. Testouri *et al.* [62] proposed a straightforward yet effective formulation by approximating obstacles as circles with safety margins and directly embedding them into the MPPI cost function. While such direct penalty formulations are effective for basic collision avoidance, they may inadvertently introduce local minima in environments with complex or nonconvex obstacles. Fuke *et al.* [60] explicitly addressed entrapment in such local minima by introducing Repulsive Potential Augmented-MPPI (RPA-MPPI), which adds an artificial repulsive potential field to the cost function to push trajectories away from trap regions. However, RPA-MPPI requires prior knowledge of the local minimum position and keeps the repulsive potential always active, causing unnecessary path deviations even without entrapment risk. They later proposed Dynamic Repulsive Potential Augmented-MPPI (DRPA-MPPI) [61] to overcome these limitations. DRPA-MPPI automatically detects imminent entrapment from the predicted trajectory of MPPI and dynamically switches between target-directed and detour-inducing costs through a virtual target that decouples the detour range from the goal distance. Fig. 5 presents the behavioral difference between RPA-MPPI and DRPA-MPPI, where Fig. 5a shows the always-active guidance field of RPA-MPPI and Fig. 5b depicts the mode-switching sequence of DRPA-MPPI from target-directed to detour-inducing and back to target-directed.

Broadening the scope from binary collision penalties and local-minimum avoidance to spatially structured risk modeling, Ardiyanto *et al.* [63] proposed Eikonal-MPPI, which augments the MPPI cost function with an Eikonal-derived

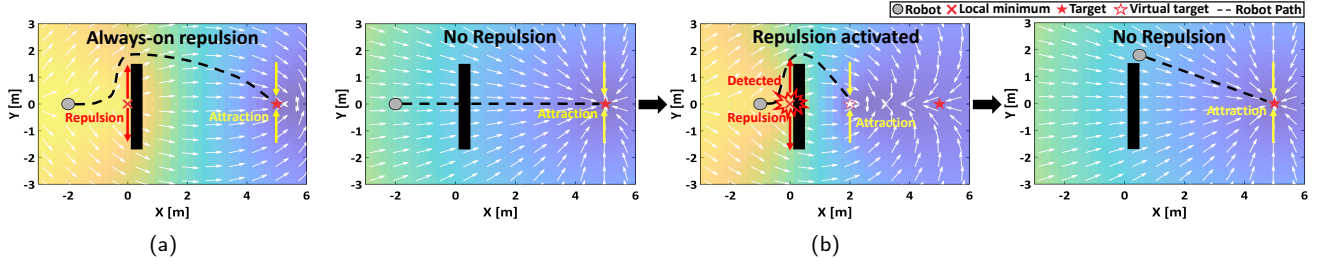


Figure 5: Comparison of operational schemes in repulsive-potential-based MPPI frameworks [60, 61]. (a) RPA-MPPI [60], where a fixed repulsive potential is always applied to push the robot away from the local minimum. (b) DRPA-MPPI [61], which activates the repulsive potential only when a local-minimum condition is detected and redirects the robot towards a dynamically generated virtual target. The sequence shows the mode transition from target-directed to detour-inducing and back to target-directed after passing the local minimum.

arrival-time field computed from a spatial risk map. This compact global cost representation biases sampled trajectories towards smoother, lower-risk regions while preserving computational efficiency. In a different but conceptually related direction, Zhai *et al.* [64] proposed Perception-Aware MPPI (PA-MPPI), which defines collision and perception-aware cost terms on an online occupancy map. Rather than providing global risk guidance, PA-MPPI favors safe viewpoints that reveal goal-directed unknown regions, thereby enabling reference-free navigation and online map expansion in previously unknown environments.

Beyond spatial constraints, penalty-based formulations have also been extended to more general constraint structures and long-horizon reasoning. In this context, Crestaz *et al.* [65] proposed Temporal-Difference Constraint-Discounted MPPI (TD-CD-MPPI), which handles constraints by modulating the discount factor of sampled trajectories according to their degree of constraint violation and augmenting the return with a terminal value function learned via temporal-difference updates. Through this constraint-discounted return formulation, trajectories that incur constraint violations are continuously down-weighted over the prediction horizon, enabling soft constraint handling without explicit barrier functions, projections, or feasibility filtering.

3.2.2. Probabilistic Constraints

Beyond penalty-based formulations, probabilistic constraint handling aims to regulate safety under uncertainty by explicitly considering stochastic variability in the system and environment. Unlike deterministic constraints, these approaches reason about risk at the distributional level and seek to control either the likelihood or the severity of constraint violations. A variety of such formulations have been explored in the MPPI literature, ranging from implicit distribution shaping to explicit probabilistic constraints and risk-sensitive objectives.

An early line of work regulates probabilistic risk implicitly by shaping the state distribution, without introducing explicit chance constraints. Yin *et al.* [66] proposed Covariance-Controlled MPPI (CC-MPPI), which augments standard MPPI with an explicit terminal covariance bound.

Using covariance steering on a linearized auxiliary dynamics model, CC-MPPI shapes the dispersion of sampled trajectories and improves robustness under uncertainty, while retaining the sampling-based information-theoretic update structure of MPPI. Although no explicit probability constraint is imposed, reducing terminal covariance can indirectly mitigate the likelihood of large stochastic deviations.

The above implicit regulation naturally motivates the explicit formulation of probabilistic safety through chance constraints. A common formalization enforces that a state-dependent constraint function $g(x)$ is satisfied with high probability:

$$\Pr [g(\mathbf{x}_k) \leq 0, \forall k \in [0, N]] \geq 1 - \delta, \quad (70)$$

where δ specifies the admissible level of risk. This formulation enables a direct trade-off between safety and performance but also introduces significant computational challenges. A related line of work further considers scenarios in which model uncertainty must be learned directly from data. Trivedi *et al.* [67] proposed chance-constrained Gaussian Process MPPI (GP-MPPI), in which safety is formulated through chance constraints but implemented via deterministic covariance-based constraint tightening derived from GP uncertainty. By adapting safety margins to the learned model uncertainty and leveraging GPU-parallelized sampling, the method achieves real-time, uncertainty-aware navigation for high-speed skid-steer robots across varying terrains. Although the chance constraint appears explicitly in the problem formulation, the implementation avoids direct probability evaluation by using deterministic uncertainty margins.

In contrast, Balci *et al.* [68] introduced Constrained Covariance Steering MPPI (CCSMPP), which realizes chance-constrained safety explicitly under linear Gaussian assumptions. By reformulating chance constraints as deterministic convex conditions and enforcing them around an MPPI-generated nominal trajectory, CCSMPP provides high-probability safety guarantees. However, the method is restricted to stochastic linear systems, which limits its applicability to general nonlinear dynamics. To overcome this limitation and extend chance-constrained MPPI to nonlinear systems, Yin *et al.* [69] proposed the Belief-Space

Stochastic MPPI (BSS-MPPI). The method avoids explicit system linearization by leveraging MC forward simulation to propagate the mean and covariance of the state under the original nonlinear dynamics. To make probabilistic safety evaluation computationally tractable, it approximates each chance constraint in belief space using a local first-order expansion together with a conservative covariance back-off.

Similarly, Mohamed *et al.* [70] introduced the Chance-Constrained Unscented MPPI (C²U-MPPI), which explicitly incorporates collision-avoidance chance constraints into a sampling-based MPC framework for nonlinear systems. By propagating uncertainty through the nonlinear dynamics using the Unscented Transform (UT) and reformulating probabilistic collision checking as a deterministic Mahalanobis-distance-based condition with a dynamically updated threshold, the method achieves computationally tractable chance-constrained control under uncertainty. Building on C²U-MPPI, Parwana *et al.* [71] extended MPPI to stochastic hybrid systems by combining Expansion-Compression Unscented Transform (ECUT)-based uncertainty propagation with sigma-point-level mode switching. They formulate safety using probabilistic collision-avoidance constraints, but solve the problem approximately through a risk-aware surrogate cost rather than directly enforcing those constraints within MPPI.

Instead of relying on analytic or deterministic approximations of chance constraints, an alternative is to estimate collision probability directly via sampling. Trevisan *et al.* [72] proposed Dynamic Risk-Aware MPPI (DRA-MPPI), which uses an efficient parallel MC scheme to approximate the joint collision probability of candidate trajectories. This estimate is then used to penalize or effectively suppress trajectories that violate a prescribed risk threshold. The method can also accommodate potentially non-Gaussian obstacle predictions while retaining real-time performance. As an alternative to online probability computation, Ezeji *et al.* [73] proposed Bayesian-Constraints MPPI (BC-MPPI), which employs an offline-trained Bayesian neural network surrogate to estimate the feasibility probability for each sampled rollout. This probability directly multiplies the standard MPPI importance weight, automatically suppressing unsafe trajectories. However, the surrogate must be retrained when the robot platform or environment configuration changes, limiting its immediate transferability to new deployment scenarios.

Finally, rather than bounding violation probability itself, a complementary class of methods quantifies the severity of rare events using coherent risk measures. A representative example is the Conditional Value-at-Risk (CVaR), which measures the expected cost over the worst $(1 - \alpha)$ tail of the outcome distribution. Generally defined for a random cost variable Z , it is expressed as:

$$\text{CVaR}_\alpha(Z) = \mathbb{E}[Z \mid Z \geq \text{VaR}_\alpha(Z)], \quad (71)$$

where $\text{VaR}_\alpha(Z)$ denotes the Value-at-Risk at confidence level α . Unlike chance constraints, CVaR accounts not only for the likelihood of adverse outcomes but also for their

severity. Along this direction, Yin *et al.* [74] proposed Risk-Aware MPPI (RA-MPPI), which accounts for tail risk under stochastic dynamical disturbances by estimating CVaR from disturbed trajectory rollouts and incorporating a CVaR-based penalty into the MPPI framework.

3.2.3. Safety Filtering

While the previous subsection focuses on regulating safety in a probabilistic or risk-aware manner, safety filtering aims to enforce hard safety constraints directly at the control input level. The most representative methodology for implementing a safety filter is the use of Control Barrier Functions (CBFs). A CBF defines a safe set within the state space and specifies conditions on the control input to prevent the system from exiting this set. Assuming the system dynamics are affine with respect to the control input, such as $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{G}(\mathbf{x})\mathbf{u}$, and the safe set is defined as $C = \{\mathbf{x} \in \mathbb{R}^n \mid h(\mathbf{x}) \geq 0\}$, the CBF condition to ensure forward invariance is expressed as:

$$\frac{\partial h}{\partial \mathbf{x}}(\mathbf{f}(\mathbf{x}) + \mathbf{G}(\mathbf{x})\mathbf{u}) \geq -\alpha(h(\mathbf{x})), \quad (72)$$

where α is an extended class $\mathcal{K}$ function that specifies the decay rate of the safety margin. This inequality becomes a linear constraint on the control input $\mathbf{u}$ and can be efficiently integrated into optimization-based control frameworks. While CBFs naturally fit into Quadratic Programming (QP) formulations used in many MPC schemes, their integration with sampling-based methods like MPPI is less straightforward. Since MPPI generates control sequences through stochastic sampling rather than solving an optimization problem with explicit constraints, incorporating hard CBF constraints requires alternative strategies that either modify the generated samples to satisfy safety conditions or reshape the sampling process itself to produce only safe trajectories.

Among such approaches, Yin *et al.* proposed Shield-MPPI [75], which incorporates an approximate discrete-time CBF violation penalty into the cost evaluation of sampled trajectories, thereby discouraging unsafe rollouts. When the MPPI-generated control still violates or is likely to violate the safety constraint, the method further applies a local gradient-based repair step to reduce CBF violations and improve safety. However, since the safety mechanism in Shield-MPPI relies on finite-horizon penalty shaping together with a local repair step, its ability to explicitly address safety beyond the prediction horizon may remain limited in more challenging scenarios. To address these limitations, the same research group subsequently developed Neural Shield-Variational Inference MPC (NS-VIMPC) [76]. Rather than relying solely on finite-horizon penalty shaping and local repair, this approach learns an approximate discrete-time CBF using a neural network and combines it with a resampling-based rollout strategy that improves the sampling process by resampling unsafe rollouts from safe ones. By integrating

learned barrier functions with sampling-aware safety enforcement, NS-VIMPC more effectively addresses safety beyond the prediction horizon while maintaining high sample efficiency.

Extending MPPI with stability-oriented trajectory shaping and execution-time safety filtering, Zhao *et al.* [77] proposed Lyapunov Function augmented-MPPI (LF-MPPI). In LF-MPPI, Control Lyapunov Function (CLF)-based terms are incorporated into the trajectory cost to downweight sampled trajectories with less favorable tracking and stability properties during rollout. After the standard MPPI update produces a nominal control input u_t^* as in (18), a QP is solved at the execution stage to obtain a control input that remains close to u_t^* while satisfying CBF constraints. Thus, LF-MPPI combines rollout-stage stability shaping with execution-time hard safety filtering. While this hybrid formulation improves stability and safety in the reported simulations, it requires solving a QP at every control step, increasing online computational complexity and motivating subsequent approaches based on more direct closed-form safety filtering mechanisms. The methods discussed above focus on penalizing or repairing potentially unsafe trajectories after generation. In contrast, proactive approaches enforce safety by construction, modifying the realized control input so that only safe trajectories are sampled and executed.

Along this direction, Rabiee and Hoagg [78] proposed Guaranteed-Safe MPPI (GS-MPPI), which constructs a composite CBF-based closed-form safety filter and embeds it directly into the planning dynamics used by MPPI. Specifically, the method combines multiple safety constraints into a single composite barrier and maps each desired control input to a guaranteed-safe control through a minimum-intervention law. By incorporating this closed-form filter into the system dynamics, GS-MPPI ensures that only safe trajectories are generated during sampling, thereby improving sample efficiency while maintaining safety by construction. For stochastic systems, Tao *et al.* [79] proposed SCBF-MPPI, which incorporates Stochastic Control Barrier Functions (SCBFs) into MPPI by constraining the sampling distribution itself. Rather than relying on post-sampling correction or execution-time filtering, the method adjusts the mean and covariance of the sampling distribution so that sampled control perturbations satisfy SCBF-based chance constraints. In this way, SCBF-MPPI restricts exploration to probabilistically safe regions of the control space, improving sample efficiency while enforcing safety in a probabilistic sense.

Beyond the CBF paradigm, Borquez *et al.* [80] developed DualGuard-MPPI, which uses Hamilton–Jacobi reachability analysis to generate provably safe rollouts during the sampling process while also applying a least-restrictive safety filter to the final control input. This approach provides safety guarantees grounded in differential-game-based reachability rather than barrier-function inequalities. Wang *et al.* [81] proposed DBaS-Log-MPPI, which embeds safety into the system dynamics via discrete barrier states and incorporates the resulting barrier-state cost into the MPPI

objective. By combining this safety-embedded formulation with adaptive exploration and log-enhanced sampling, the method improves feasibility and trajectory optimization performance in cluttered environments.

3.2.4. Constraint-Aware Sampling

The standard Gaussian sampling strategy employed in MPPI does not explicitly account for control or environmental constraints, leading to substantial sampling inefficiency when a large portion of generated trajectories violate dynamic limits or collide with obstacles. Since such infeasible samples receive negligible importance weights, the quality of the resulting update can be significantly degraded, particularly in tightly constrained environments.

To address this limitation, a line of recent studies has introduced post-sampling correction mechanisms that explicitly enforce feasibility by modifying sampled control sequences before trajectory rollouts. Within this category, Andrejev *et al.* [82] proposed π -MPPI, which integrates a projection-based filter to enforce control magnitude and higher-order derivative constraints. Each sampled control sequence is minimally perturbed to satisfy predefined bounds through a quadratic programming-based projection step. This projection process is efficiently parallelized on GPUs using a customized Alternating Direction Method of Multipliers (ADMM) solver with neural-network-based warm starts. However, since the correction operates purely in the control-input space, state-level feasibility, such as collision avoidance, is not directly enforced.

To enhance state-level feasibility, Miura *et al.* [83] proposed Sparse Control Points MPPI (SCP-MPPI), which combines spline interpolation with sample transport. Instead of sampling full control sequences, SCP-MPPI samples sparse control points and reconstructs smooth trajectories via cubic spline interpolation. To compensate for the reduced temporal resolution and mitigate collision risks, SVGD is applied to iteratively transport the sparse control points towards collision-free and low-cost regions while preserving diversity through repulsive kernel interactions.

Adopting a different correction paradigm, Park *et al.* [84] proposed Constrained Sampling Cluster-MPPI (CSC-MPPI), which combines primal–dual constraint refinement with density-based clustering. First, each sampled control sequence is iteratively adjusted via primal–dual updates until the KKT conditions are approximately satisfied, thereby shifting infeasible samples into the feasible region. Then, DBSCAN is applied to the feasible trajectories, and a representative control input is computed for each cluster. Finally, the lowest-cost representative is selected for execution. This clustering-based selection mitigates the risk of constraint violation that can arise when heterogeneous feasible trajectories are directly combined through weighted averaging. A visual comparison of SCP-MPPI [83] and CSC-MPPI [84] is provided in Fig. 6.

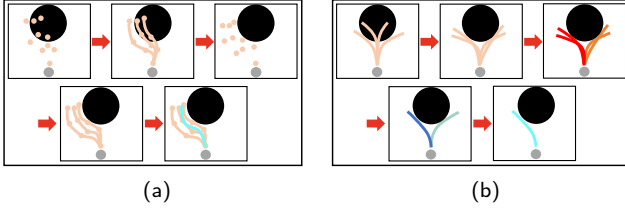


Figure 6: Methodological comparison of two post-sampling correction strategies. (a) SCP-MPPI [83] transports sampled trajectories towards low-cost regions using SVGD, compensating for the low predictive resolution of sparse control points. (b) CSC-MPPI [84] moves samples towards feasible regions via primal-dual gradient updates, enforcing KKT-based constraints before selecting representative feasible trajectories.

3.3. Framework Design

Recent studies have extended the MPPI control framework by combining it with various forms of optimization [85], learning [86], and inference algorithms [87] to enhance convergence and adaptability. In addition, evolutionary optimization approaches, such as the MPPI-Genetic Algorithm (MPPI-GA) [88], have demonstrated the potential of integrating MPPI with heuristic search frameworks. While such algorithmic combinations are common across stochastic optimal control frameworks, a particularly active research trend focuses on structural integration, where MPPI is embedded within broader control architectures. These architecture-oriented designs aim to leverage the strengths of MPPI in handling nonlinear dynamics while addressing system-level concerns such as scalability and long-horizon planning. The following subsections review representative works that adopt this structural perspective, organized into two lines of integration: hierarchical integration and multi-MPPI integration.

3.3.1. Hierarchical Integration

Hierarchical integration incorporates MPPI within multi-level control architectures that separate high-level planning from low-level execution, addressing the computational complexity of long-horizon trajectory optimization through problem decomposition. MPPI can operate at either level depending on system requirements, serving as a local optimizer that refines coarse plans from higher-level modules or as a trajectory generator tracked by dedicated low-level controllers. This architectural flexibility enables hierarchical systems to balance global coherence with local reactivity while maintaining scalability and modularity.

A primary category within this architecture employs MPPI as a local trajectory refiner that executes or optimizes a coarse plan generated by a higher-level module. For instance, Li *et al.* [89] proposed a Global Path Guided MPPI framework, where a high-level global planner generates a global path or waypoint sequence and MPPI performs local optimization to track it. This integration helps mitigate the tendency of vanilla MPPI to get trapped in local optima in large-scale or cluttered environments, although its effectiveness depends strongly on the quality of the global planner.

Similarly, Mohamed *et al.* [90] proposed Gaussian Process-guided MPPI (GP-MPPI), where Sparse Gaussian Processes (SGPs) are leveraged to identify informative subgoals from local uncertainty estimates. In this architecture, the GP acts as an upper-level guidance module, while MPPI handles feasible local trajectory optimization and execution.

Along the same lines, Zhang *et al.* [91] developed the Risk-Aware Parallel Planner (RAPA-Planner), a hierarchical planning framework consisting of an efficient front-end and a back-end Risk-Aware MPPI (RA-MPPI) optimizer. In this framework, the front-end constructs a topology graph to generate guiding trajectories, which are then refined by the back-end RA-MPPI optimizer. The latter incorporates the CVaR, as defined in (71), to quantify and mitigate risk, thereby enhancing safety and robustness. Furthermore, hierarchical integration can be extended to the task-motion level. Zhang *et al.* [92] proposed a framework combining an Active Inference Planner (AIP) with a novel Multi-Modal MPPI (M3P2I) controller. In this framework, the task-level AIP generates alternative symbolic action plans, which are then translated into corresponding cost functions for M3P2I. The low-level controller samples across these alternatives and fuses them into a coherent motion, enabling reactive adaptation among skills such as pushing, pulling, or grasping.

Beyond single-agent settings, hierarchical MPPI architectures have also been adopted for multi-agent coordination. Jansma *et al.* [93], building on Interaction-Aware MPPI (IA-MPPI) control [94], proposed an interaction-aware hierarchical framework that integrates a learning-based trajectory prediction and local-goal extraction module with IA-MPPI. The upper level predicts the future motion of surrounding agents and infers their local goals, while the lower-level IA-MPPI performs decentralized, interaction-aware motion planning based on these goals, enabling communication-free navigation in dense multi-agent environments. In the domain of multiple Autonomous Underwater Vehicles (AUVs), Zhao *et al.* [95] proposed a three-layer hierarchical framework in which a Subproblem-Based Adaptive Coordination planner generates collision-free reference paths, a Guiding Vector Field layer smooths them into continuous trajectories, and a low-level MPPI controller optimizes control inputs under full nonlinear AUV dynamics and environmental disturbances.

A second category of hierarchical design reverses this relationship, employing MPPI as a high-level trajectory generator whose output is tracked by a separate, often faster, low-level controller. This approach leverages MPPI for its planning strengths while delegating real-time safety and robustness to a dedicated follower. Vasilopoulos *et al.* [96] introduced the Reactive Action and Motion Planner (RAMP), a hierarchical framework in which an MPPI-based trajectory generator proposes reference trajectories that are asynchronously tracked by a low-level vector-field follower. This design improves safety and real-time reactivity, though

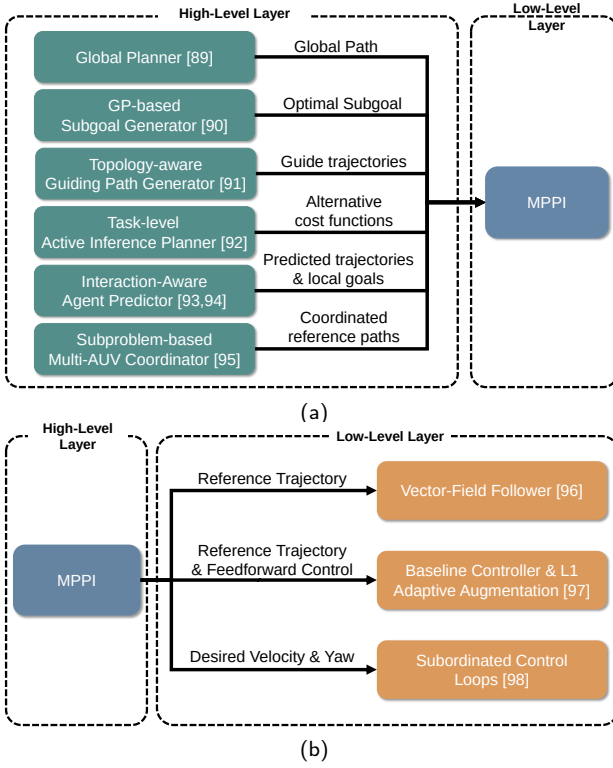


Figure 7: Overview of hierarchical integration architectures. (a) MPPI serves as a low-level local optimizer that refines coarse plans or guides provided by high-level planners. (b) MPPI functions as a high-level trajectory generator, providing reference states to dedicated low-level tracking controllers.

the executed motion may deviate from the planned trajectory due to local obstacle avoidance and tracking behavior, leading to a trade-off between trajectory fidelity and reactive safety. Related hierarchical strategies further combine such planner-tracker decompositions with adaptive feedback mechanisms. Pravitra *et al.* [97] proposed an L1-Adaptive MPPI architecture, in which MPPI acts as a non-linear MPC module that generates an optimized trajectory and feedforward control sequence, while a baseline tracking controller with L1 adaptive augmentation compensates for tracking errors and model mismatch in real time. Homburger *et al.* [98] further investigated subordinated control loops with MPPI by explicitly comparing MPPI applied directly to the nonlinear plant with MPPI operating on a simplified model of the low-level closed-loop dynamics. Their analysis showed that such layering improves robustness to disturbances and model-plant mismatch by providing a more predictable effective model for the high-level MPPI controller. To visualize the distinction between these two hierarchical paradigms, we present a schematic overview in Fig. 7, highlighting the input-output relationships specific to each configuration.

3.3.2. Multi-MPPI Integration

Beyond these hierarchical integrations between MPPI and external planners or controllers, recent research extends the MPPI framework by employing multiple MPPI instances

within a unified control structure to achieve more diverse behaviors and improved overall performance. For instance, Zhou *et al.* [99] developed the Parallel MPPI (PMPPI) architecture, in which multiple planners with different strategies, such as goal seeking and obstacle avoidance, run concurrently. A separate Judge strategy evaluates their trajectories under a shared criterion and adaptively fuses their outputs into a single control policy. This structural decomposition enables diverse behaviors to be coordinated within a unified MPPI framework, improving exploration capability and robustness in dynamic environments while maintaining near-real-time computational performance through parallelized execution.

Similarly, Raisi *et al.* [100] proposed Planner-Estimator MPPI (PE-MPPI), a fault-tolerant control framework for free-flying space robots. The Planner MPPI computes control actions using an onboard nominal model, while the Estimator MPPI updates the model parameters when the discrepancy between the real system and onboard model responses exceeds a threshold. This planner-estimator structure improves robustness to structured parameter uncertainties and actuator failures by adapting the planner's dynamics model online and preserving control performance. Jung *et al.* [101] further developed this multi-instance idea into a nested framework called Contingency-MPPI. In this architecture, an outer nominal MPPI optimizes a nominal trajectory, while an inner contingency MPPI is executed along each candidate nominal rollout to verify the existence of a feasible backup path to a safe zone within a finite horizon. If the inner planner fails to find such a contingency for any state along the rollout, that trajectory is discarded by assigning it infinite cost. This nested design enforces contingency reachability throughout the nominal plan, although it increases computational burden and therefore relies on additional seeding strategies to improve sampling efficiency.

3.4. Robustness Enhancement

Real-world robotic systems seldom operate in benign conditions. Instead, they are constantly challenged by model inaccuracies, exogenous disturbances, and incomplete or noisy information about the task environment. While the baseline MPPI algorithm has demonstrated remarkable performance in nominal conditions, its risk-neutral design renders it fragile in the face of uncertainty and disturbances. As a result, a significant body of work has focused on extending MPPI towards more robust formulations. In this subsection, we review advances along two dimensions: methods that explicitly or implicitly account for uncertainty in dynamics, costs, and sampling distributions, and methods that enhance robustness to external disturbances and environmental variability.

3.4.1. Robustness to Uncertainty

The baseline MPPI formulation relies on a nominal dynamics model and assumes idealized conditions during trajectory evaluation. In practical robotic systems, however, several forms of uncertainty inevitably arise, including parametric mismatch in the system model, stochastic

disturbances that affect state evolution, and variability in sensing or actuation. These discrepancies between the nominal model and real-world behavior have motivated many extensions that directly address these uncertainties, allowing MPPI to better handle model errors, external perturbations, and other sources of variability encountered during deployment.

A primary challenge in model-based control is the discrepancy between the mathematical model and the physical system. To address parameter mismatch, Abraham *et al.* [102] extended the free-energy formulation in (5) to explicitly incorporate uncertainty over model parameters. In their approach, the system dynamics are parameterized by a distribution $p(\bar{\theta})$, and the free energy is defined as

$$F_{\lambda} = -\lambda \log \mathbb{E}_{p(\bar{\theta}), \mathbb{P}} \left[\exp \left(-\frac{1}{\lambda} S(\mathbf{V}) \right) \right]. \quad (73)$$

This expression expands the expectation in the nominal formulation to capture both uncertainty in the model parameters and the stochasticity introduced by trajectory sampling. The resulting formulation allows trajectories to be evaluated under a broader uncertainty model without requiring explicit identification of the underlying parameters. Addressing similar parametric sensitivities, Nyboe *et al.* [103] proposed a robust MPPI formulation based on embedded sensitivity tubes. They propagate state-parameter sensitivities alongside the nominal dynamics to approximate local uncertainty envelopes. These tubes are incorporated into the cost function as robustness penalties, improving control performance under parameter mismatch. Alternatively, instead of explicit parameterization, Cong *et al.* [104] proposed Recurrent MPPI (RMPPi), which employs a recurrent neural network trained under domain randomization. The LSTM model adapts online to compensate for unknown physical parameters and sim-to-real discrepancies, demonstrating that robustness can emerge from data-driven adaptation.

Closely related to parameter mismatch is the epistemic uncertainty inherent in learned dynamics models, particularly in data-sparse regions. Arruda *et al.* [105] addressed this issue by incorporating predictive uncertainty from learned forward models into the stage cost. Accordingly, the trajectory cost in (3) is reformulated to explicitly account for predictive uncertainty as

$$S(\mathbf{V}) = \phi(\mathbf{x}_T) + \sum_{t=0}^{T-1} \ell(\mathbf{x}_t, \hat{\sigma}_t), \quad (74)$$

where $\hat{\sigma}_t$ denotes the predictive uncertainty estimated by a Gaussian process or an ensemble of mixture-density networks. For instance, in a goal-tracking task, the uncertainty-aware stage cost can be written as

$$\ell(\mathbf{x}_t, \hat{\sigma}_t) = \gamma \hat{\sigma}_t + (\mathbf{x}_t - \mathbf{x}_{\text{goal}})^{\top} \mathbf{Q} (\mathbf{x}_t - \mathbf{x}_{\text{goal}}). \quad (75)$$

Similarly, Kalaria *et al.* [106] developed an uncertainty-aware MPPI that estimates ensemble-based epistemic uncertainty and incorporates it as a risk-sensitive term in the cost. Coupled with meta-learning-driven model adaptation,

this method achieves robust performance under changing dynamics and surface conditions.

Beyond the structure of the model, the nature of the stochastic process itself can also introduce additional uncertainty, particularly when the Gaussian noise assumption in (19) is violated. To address this limitation, Wang *et al.* [107] extended the baseline dynamics in (19) to a jump-diffusion system in order to capture rare but significant stochastic shocks. The resulting extended dynamics are expressed as

$$d\mathbf{x}_t = f(\mathbf{x}_t) dt + G(\mathbf{x}_t) \mathbf{u}_t dt + B(\mathbf{x}_t) d\mathbf{w}_t + H(\mathbf{x}_t) d\mathbf{P}_t, \quad (76)$$

where $H(\mathbf{x}_t) d\mathbf{P}_t$ represents discontinuous jump disturbances governed by a compound Poisson process. This generalization enables the modeling of non-Gaussian, jump-like disturbances that may arise from abrupt environmental interactions. In a related effort to handle stochastic state evolution, Mohamed *et al.* [108] proposed the Unscented MPPI (U-MPPI). This approach leverages the UT to propagate both the mean and covariance of the state distribution through nonlinear dynamics. By jointly considering uncertainty propagation and employing a risk-sensitive cost that explicitly depends on the propagated state covariance, U-MPPI achieves safer control under stochastic uncertainty.

Uncertainty also arises downstream at the execution level, where the low-level controller may fail to perfectly track the high-level plan. Higgins *et al.* [109] addressed this issue by embedding a data-informed risk term into the MPPI cost for UAV planning. In their approach, tracking errors are quantified through a data-driven estimate of trajectory mismatch that depends on vehicle speed and obstacle clearance. By penalizing trajectories with high risk, the method enables the UAV to proactively adjust its velocity near obstacles, effectively incorporating tracking-induced execution uncertainty into the planner. Finally, uncertainty also arises from the external environment and from the sampling process itself. Patrick and Bakolas [110] addressed the risk of multimodal cost landscapes and dynamic environments by introducing rollout clustering and probabilistic dynamic-obstacle costs. Their method clusters sampled rollouts to prevent the weighted average from passing through unsafe regions, and it incorporates sampled obstacle trajectories with associated probabilities into the trajectory cost. This enables the controller to anticipate stochastic obstacle behavior and generate collision-averse trajectories in dynamic environments. To consolidate the above literature, Table 3 summarizes the types of uncertainty addressed by each MPPI variant.

3.4.2. Robustness to Disturbances

While uncertainty in system parameters and models is a critical challenge, disturbances constitute an equally important factor that can severely compromise the performance of MPPI. Real-world systems are exposed to sudden shocks, structured external forces, and adversarial environmental effects that cannot be captured by Gaussian input noise alone. To address this, several extensions of MPPI have been

Table 3

Uncertainty types addressed by MPPI variants.

Method	Parametric	Process Noise	Epistemic	Execution Error	Environment
Abraham <i>et al.</i> [102]	✓				
Nyboe <i>et al.</i> [103]	✓				
Cong <i>et al.</i> [104]	✓				
Arruda <i>et al.</i> [105]	✓		✓		
Kalaria <i>et al.</i> [106]	✓		✓		
Wang <i>et al.</i> [107]		✓			
Mohamed <i>et al.</i> [108]		✓			
Higgins <i>et al.</i> [109]				✓	
Patrick & Bakolas [110]					✓

✓: Addressed

proposed that explicitly focus on disturbance rejection and robust performance.

Williams *et al.* [111] proposed a robust extension of MPPI within a tube-based model predictive control framework to explicitly handle external disturbances. In their approach, MPPI serves as a nominal controller that optimizes a control sequence for an idealized nominal system, while an ancillary tracking controller stabilizes the deviation between the nominal and the actual disturbed system. This separation helps keep the actual system close to the nominal MPPI trajectory, thereby improving robustness against disturbances and reducing the risk of divergence in sampling-based MPC. Building on this, Gandhi *et al.* [112] proposed Robust-MPPI (RMPPI), which extends the tube-based MPPI framework by incorporating augmented-state sampling, feedback-aware importance sampling, and nominal state propagation. Unlike earlier robust MPPI variants whose disturbance rejection properties were primarily supported empirically, RMPPI provides a formal characterization of disturbance rejection by deriving explicit bounds on real-system free-energy growth under suitable assumptions on the tracking controller. This moves robust MPPI beyond observed empirical behavior towards a theoretically grounded analysis of disturbance handling. To handle adversarial disturbances more rigorously, Zhang *et al.* [113] proposed Zero-Sum Game MPPI (ZSG-MPPI), which models the disturbance as an adversarial player in a Stackelberg zero-sum game and establishes an equivalence with a specific optimal control problem via Hamilton-Jacobi analysis. This formulation enables importance-sampled MPPI to compute a control policy that explicitly accounts for the corresponding adversarial disturbance in real time.

While the above approaches address disturbances at the control level, either through tube-based tracking or game-theoretic formulation, a complementary line of work instead improves the prediction dynamics themselves. Zhang *et al.* [114] proposed Disturbance-Aware MPPI (DA-MPPI), which incorporates disturbance estimates directly into the rollout model. An Extended high-order Sliding Mode Observer jointly estimates system states and lumped disturbances in finite time, and these estimates are directly incorporated into the rollout dynamics used during MPPI trajectory rollouts, allowing the controller to account for

mismatched disturbances over the receding horizon. In a similar spirit of improving the predictive model, Esfandiari *et al.* [115] proposed an adaptive framework where the prediction dynamics are corrected online. Their method augments a data-driven Gaussian Mixture Model–Gaussian Mixture Regression (GMM-GMR) model with an Extended Kalman Filter (EKF)-updated Radial Basis Function (RBF) identifier, which continuously adjusts the local dynamics used by MPPI to compensate for unknown external disturbances and environmental loads.

Liang *et al.* [116] further applied MPPI to missile guidance under maneuvering targets, varying interceptor velocity, and actuator failures. Their framework integrates a deep neural network dynamics model into MPPI and uses meta-learning to adapt this predictive model online from recent experience, so that changes induced by target maneuvers, environmental perturbations, and actuator loss-of-effectiveness are reflected directly in the rollout dynamics. By updating the learned dynamics in real time, the method achieves robust interception with desired terminal impact angles, demonstrating MPPI's applicability to safety-critical and disturbance-rich domains. Transitioning from internal faults to external environmental disturbances, Folk *et al.* [117] proposed an energy-aware MPPI for UAV navigation in urban environments where gusts act as strong exogenous disturbances. Their method augments the cost function with power consumption penalties informed by onboard LiDAR-based wind predictions, achieving safer trajectories with up to 30% lower energy consumption and fewer collisions, though performance inherently relies on the fidelity of local wind estimation.

4. Real-Time Implementation Strategies and Modular Frameworks for MPPI

MPPI control has evolved from a theoretical stochastic control formulation into a practical real-time framework, enabled by advancements in computational hardware and open-source software. With the rise of GPU acceleration and differentiable simulators, MPPI has transitioned from an isolated algorithmic controller into a shared research baseline that supports reproducibility, scalability, and flexible algorithmic development. This section reviews the evolution of MPPI implementations and introduces a modular open

Table 4

Summary of MPPI implementation approaches and their characteristics.

Implementation Approach	Compute Mechanism	Main Advantages	Main Limitations
Pure CPU (Section 4.1.1)	Sequential CPU execution	Low cost, Easy prototyping, Embedded compatibility	Limited parallelism, Low frequency, Poor scalability
GPU/CUDA (Section 4.1.2)	Custom CUDA kernels	Thread-level parallelism, Manual memory optimization	Hardware dependence, High development complexity
High-level GPU frameworks (Section 4.1.3)	Tensor vectorization	Modularity, Autodiff, Learning integration	Abstraction overhead, Reduced low-level control
Physics simulator integration (Section 4.1.4)	Parallel physics stepping	Realistic contact modeling, Physical validity	High computational cost, Simulator fidelity limits

framework designed to facilitate rapid research and real-time deployment.

4.1. Evolution of MPPI Implementations

At each control cycle, MPPI generates thousands of control sequence samples and propagates them through system dynamics to evaluate the expected cost. Because this procedure is computationally intensive, the historical development of MPPI implementations can be viewed as a continuous pursuit of greater numerical efficiency and real-time capability. A concise overview of this evolution is summarized in Table 4, which organizes the major implementation approaches ranging from pure Central Processing Unit (CPU) execution to GPU/CUDA kernels, high-level GPU frameworks, and physics-simulator integration, and highlights their computational mechanisms, advantages, and limitations.

4.1.1. Sequential Rollout and Early CPU Implementations

The original MPPI formulation by Williams *et al.* [13] was implemented on GPUs from the outset to exploit massively parallel rollouts. Many follow-up studies, however, were carried out on CPUs using sequential rollout simulations, particularly in embedded deployments and early prototyping stages where dedicated hardware was unavailable or unnecessary. While CPU-based implementations provided a deterministic execution environment and mature debugging tools, sequential rollouts caused the total computational cost to scale linearly with the number of samples, limiting control frequencies to a few hertz. High-dimensional or fast-dynamics systems often required simplifying the dynamics model or reducing the sample count, which degraded control performance and reaffirmed MPPI's intrinsic dependence on large-scale parallelization.

4.1.2. Parallel Sampling on GPU and CUDA Acceleration

The introduction of GPUs enabled MPPI to fully exploit the parallel structure of its sampling process. Since each rollout is independent, the sampling procedure mapped naturally onto GPU thread-level parallelism, allowing thousands of trajectories to be propagated simultaneously. Through CUDA kernels, MPPI achieved control frequencies of several hundred hertz at high sampling resolution [118], evolving into a deployable controller for high-speed robotic

systems. However, CUDA-level implementations required manual management of GPU memory, thread scheduling, and kernel fusion, which substantially increased development overhead and motivated the search for higher-level abstractions that preserve performance while improving modularity.

4.1.3. High-Level GPU Computing Frameworks

The emergence of high-level GPU computing and differentiable programming frameworks such as JAX and PyTorch marked an important transition in MPPI development [16, 84]. By removing the need for low-level GPU programming while preserving efficient parallel execution, automatic differentiation, Just-In-Time (JIT) compilation, and large-scale vectorization, these frameworks allowed MPPI computations to be expressed as vectorized operations applied to trajectory batches, increasing modularity and simplifying performance optimization.

JAX adopts a functional programming style with a compilation-based execution model, representing MPPI computations as pure and composable functions that admit extensive compiler optimizations and predictable execution. While advantageous for large-scale stochastic control requiring reproducibility and efficiency, the compilation workflow introduces overhead when model structures or control logic must change at runtime. PyTorch instead follows a dynamic computation graph philosophy that supports intuitive debugging, rapid prototyping, and seamless integration with deep learning ecosystems, allowing MPPI implementations to interact effectively with neural policy networks and data-driven cost models.

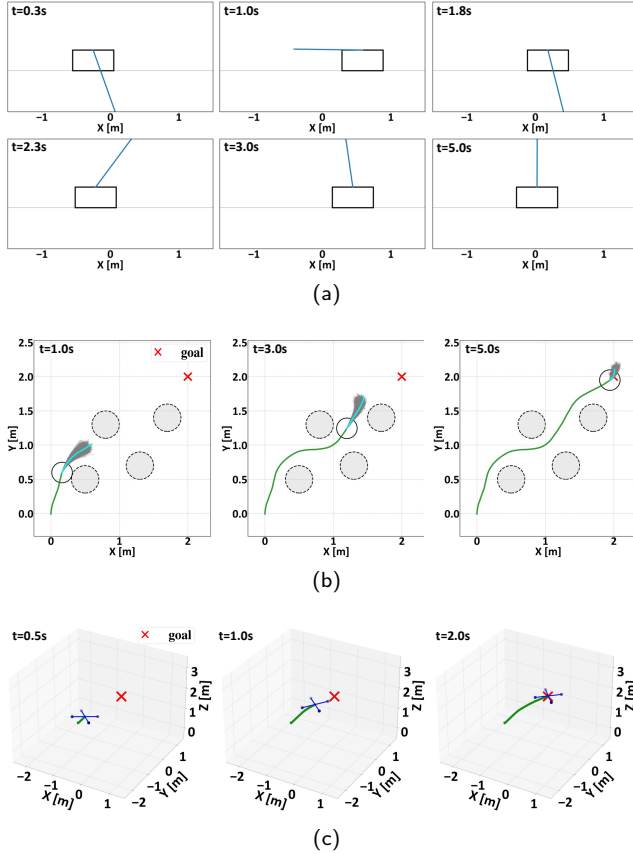
4.1.4. Integration with Physics Simulators

The recent generation of MPPI frameworks extends beyond analytical or kinematic models by directly coupling with high-fidelity physics simulators such as MuJoCo, Isaac Gym, and MJX [65, 119, 120, 121]. Through these environments, each sampled trajectory can be propagated under physically realistic conditions that include contact dynamics, frictional interactions, and external disturbances. This integration allows the controller to evaluate not only smooth trajectory optimization but also the feasibility of actions in contact-rich or dynamic environments where nonlinearity and discontinuity play a dominant role. By incorporating real-time physics feedback, MPPI can estimate quantities such as contact forces, impact impulses, and joint constraints

Table 5

Modular structure of the baseline MPPI framework and examples of researcher extensions.

Module	Function	Extensibility
Sampling Module	Generates stochastic control perturbations (noise) for trajectory rollouts	Implement various distributions and learned sampling priors
Dynamics Module	Propagates state trajectories forward in time using analytical or simulator-based models	Incorporate various robot dynamics, perform parallel physics simulation for rollouts, or replace with learned neural dynamics
Cost Module	Computes state-dependent stage costs and terminal costs for performance evaluation	Design custom objectives such as collision avoidance penalties, safety barrier functions, or energy efficiency terms
Logging / Visualization Module	Records simulation data, cost convergence trends, and control performance	Use MuJoCo viewer for real-time visual feedback or Python-based plotting tools for post-analysis


Figure 8: Application of the proposed modular framework. (a) Cart-Pole. (b) Mobile robot. (c) Quadrotor.

during each rollout, leading to more robust and physically grounded control performance. However, the computational cost of this approach is significantly higher than that of purely mathematical cost evaluations, since the underlying physics engines must solve differential contact constraints and collision responses at every simulation step.

4.2. Modular Open Framework

To support extensible research and reproducible evaluation, the proposed open-source MPPI framework provides a minimal yet flexible base implementation rather than a fully featured control suite. Fig. 8 presents three base environments included within the framework for algorithm

development and experimentation, which consist of a Cart-Pole system, a two-dimensional mobile robot, and a three-dimensional quadrotor.

The framework is organized around four core modules: a sampling module for generating stochastic control perturbations, a dynamics module for propagating state trajectories through analytical or simulator-based models, a cost module for evaluating stage and terminal costs, and a logging and visualization module for recording simulation data and performance statistics. Table 5 summarizes these modules and outlines extensibility options that allow researchers to modify or replace individual components without affecting the overall MPPI loop. Researchers may introduce new sampling distributions by subclassing the sampling interface and redefining the rollout generation routine, apply task-specific dynamics models by extending the dynamics module, or incorporate additional cost terms and hierarchical updates through custom cost modules. This modular organization supports the integration of new algorithmic ideas while preserving compatibility with the core MPPI structure.

To facilitate integration with robotic systems, the framework includes a lightweight example interface for ROS 2 Humble. In this example setup, a ROS 2 node subscribes to robot state topics and publishes control commands from the MPPI controller in real time. Fig. 9 demonstrates this capability by applying the framework, coupled with a MuJoCo physics simulator, to trajectory tracking with a seven-degree-of-freedom manipulator. We note that the Nav2 navigation stack already provides a CPU-based MPPI local controller tailored for mobile robot navigation [122]. For purely navigation-oriented applications, practitioners may prefer that mature implementation. Our framework targets a complementary scope and offers a domain-independent, GPU-accelerated MPPI structure applicable to manipulators, legged robots, aerial vehicles, and other platforms whose dynamics extend beyond planar navigation. Researchers may adapt or extend the provided ROS 2 interface by adding sensory inputs, custom estimators, or coordination nodes to fit their specific platform requirements.

4.2.1. Evaluation of the Framework's Modular Structure

To validate the modularity of the proposed framework, we evaluate the baseline implementation on the Cart-Pole stabilization task while swapping the sampling module

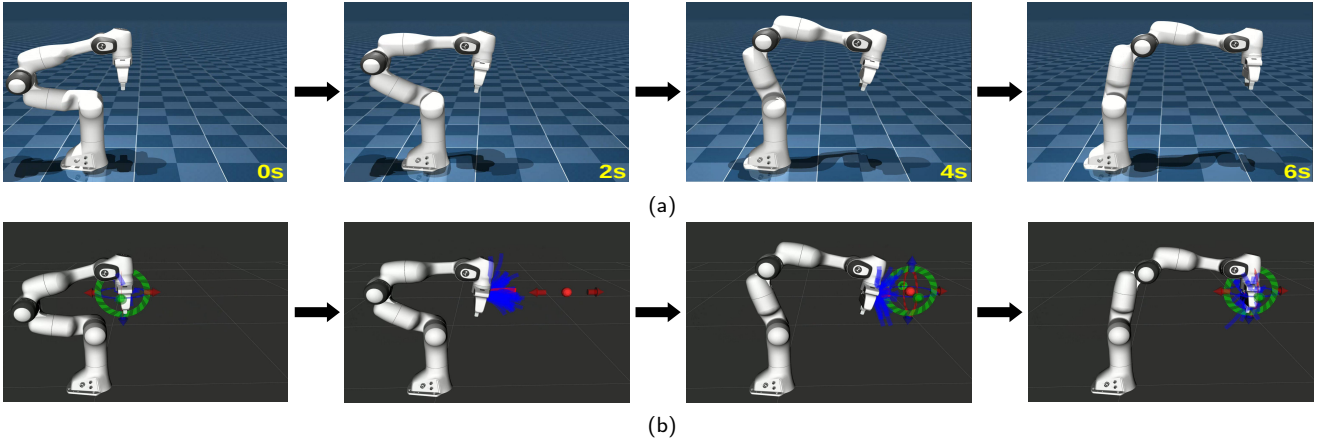


Figure 9: Demonstration of the proposed framework integrated with ROS 2 Humble and MuJoCo for trajectory tracking of a seven-degree-of-freedom manipulator. (a) Sequential snapshots from the MuJoCo simulation. (b) Corresponding RViz2 visualizations, where blue lines represent sampled trajectories and the red line denotes the optimal trajectory computed by the MPPI controller.

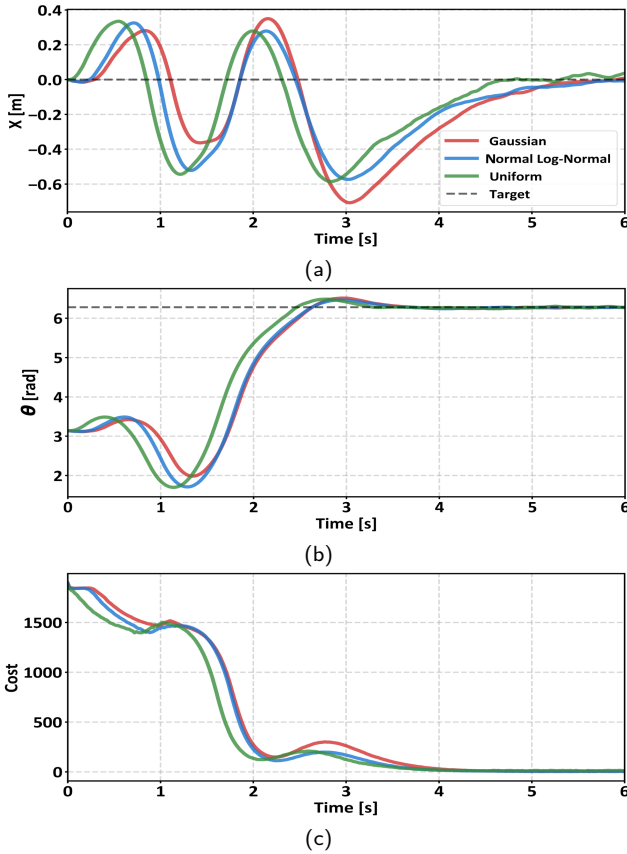


Figure 10: Optimization results for the Cart-Pole task utilizing interchangeable sampling modules within the proposed framework (Gaussian [27], NLN [47], and Uniform [48]). (a) Cart position x . (b) Pole angle θ . (c) Cost convergence history.

among Gaussian, NLN, and Uniform noise. Each strategy is integrated into the same MPPI controller without modifying any other component, demonstrating that sampling distributions can be interchanged seamlessly. As shown in Fig. 10, the controller successfully stabilizes the system across all

sampling types, with only minor differences in transient response due to varying exploration characteristics.

To further demonstrate extensibility, we conduct an additional experiment focusing on control smoothness. Fig. 11 compares six methods on the same task, namely vanilla MPPI, two filter-based baselines (Moving Average Filter (MAF) and Savitzky–Golay Filter (SGF)) applied before cost evaluation, and three smooth-sampling variants (TC-MPPI [41], Low Frequency-MPPI [42], and SMPPI [58]). The effective action standard deviation entering the dynamics is matched to approximately 3.0N across all methods (Fig. 11c), with the initial transient attributed to actuator force clamping. The planned action variation on a log scale is defined as

$$\mathcal{J}_{\text{var}} = \sum_{t=1}^{T-1} \|u_t^* - u_{t-1}^*\|^2. \quad (77)$$

As shown in Fig. 11b, the three smooth-sampling variants produce notably smoother planned sequences than both vanilla MPPI and the filter-based baselines. Together with Fig. 10, these results confirm that the framework provides a common environment for comparing algorithmic components under consistent conditions.

4.2.2. Significance of the Framework for Research

The proposed framework is not intended as a full benchmark suite but as a minimal and reproducible baseline that supports research and development activities. It offers a transparent implementation of the core MPPI algorithm together with a consistent structural template that researchers can extend or modify according to their objectives. This modular design enables systematic comparison of algorithmic variations while reducing the engineering effort required to construct low-level control infrastructure.

Because the framework maintains standardized interfaces across the modules, new ideas such as non-Gaussian sampling strategies, neural or learned dynamics models, and cost formulations informed by high-fidelity physics engines

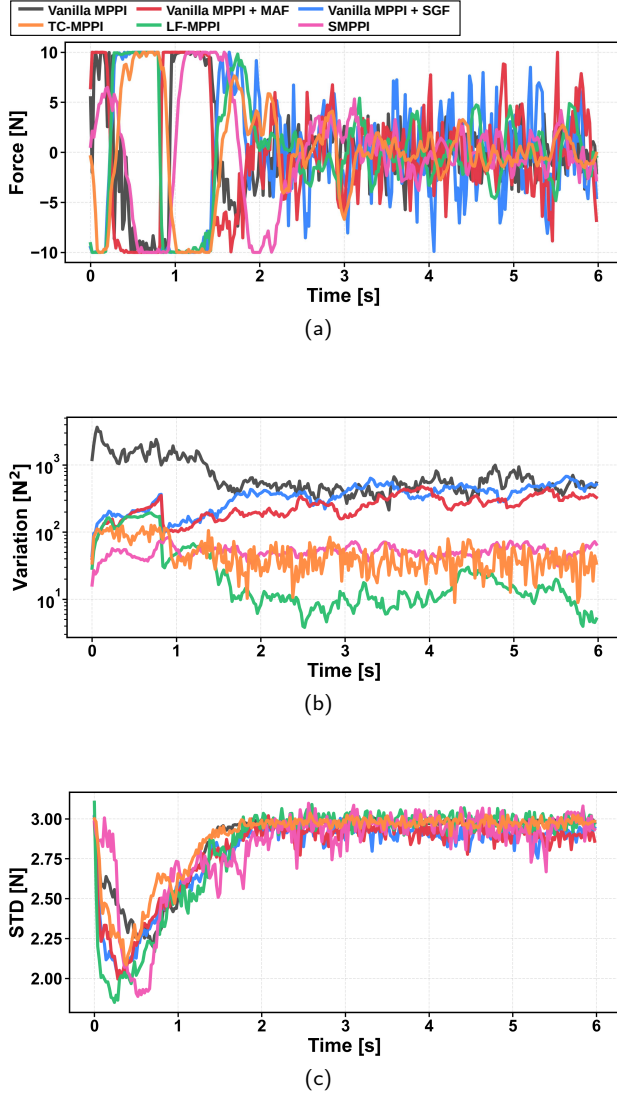


Figure 11: Comparison of control smoothness for vanilla MPPI, filter-based baselines (MAF: Moving Average Filter, SGF: Savitzky–Golay Filter), and smooth-sampling variants (TC-MPPI [41], Low Frequency-MPPI [42], SMPPI [58]) on the Cart-Pole task. (a) Executed control input. (b) Planned action variation (log scale). (c) Sampled action standard deviation.

like MJX can be evaluated within the same control loop. This consistency allows researchers to isolate the effects of individual design choices without interference from environmental or implementation differences. Through this modular yet minimal composition, the framework serves as a shared experimental foundation for developing and validating advanced MPPI variants. Rather than enforcing a single optimal configuration, it promotes transparent comparison, reproducibility, and cumulative progress in MPPI.

5. Conclusion

This paper presented a comprehensive survey of MPPI control. On the theoretical side, the information-theoretic

and classical path integral derivations were shown to be mathematically equivalent under the noise–control matching condition, the identification $\Sigma = \lambda R^{-1}$, and the invertibility of G_c , with the equivalence established through three correspondences: the objective functions, the sampling distributions, and the augmented cost terms. The comparative analysis further clarified how MPPI relates to the open-loop, closed-loop, and deterministic formulations of stochastic optimal control. On the methodological side, previously fragmented research was reorganized into four themes: sampling strategies, constraint handling, framework design, and robustness enhancement. This classification clarified the strengths and limitations of each direction and their structural relationships. On the implementation side, a modular and extensible MPPI framework was developed to support reproducible experimentation and rapid validation of algorithmic variants, serving as a practical benchmark for developing, comparing, and deploying MPPI-based controllers in real robotic systems.

5.1. Open Challenges and Future Directions

Despite the rapid progress reviewed in this survey, several fundamental limitations remain open. First, MPPI does not currently come with a general closed-loop stability theorem analogous to those available for stabilizing MPC with terminal ingredients [123, 124, 125]. A possible route towards formal stability analysis is to augment MPPI with a feasibility-preserving warm-start mechanism and a cost-improvement acceptance rule, then study whether the resulting controller admits a Lyapunov-type decrease. However, this is nontrivial for two reasons. Standard MPPI lacks the terminal ingredients (e.g., terminal cost, terminal set, and terminal control law) on which suboptimal MPC stability proofs rely, and designing such ingredients for sampling-based controllers on general nonlinear systems is an open problem. Moreover, the importance-weighted update is a soft averaging operation over stochastic samples and does not deterministically guarantee cost improvement at every replanning step, unlike the hard cost non-increase rule assumed in suboptimal MPC.

Second, sampling efficiency fundamentally degrades as the control dimensionality m and planning horizon T increase, since the probability of discovering low-cost trajectories diminishes exponentially in the trajectory space $\mathbb{R}^{m \times T}$. This limitation becomes particularly acute in high-dimensional systems such as humanoid whole-body control, where naive sampling yields nearly no dynamically feasible trajectories within a practical sample budget. While the sampling strategies reviewed in Section 3.1 partially alleviate this issue, they primarily reshape the exploration within the original actuation space without fundamentally reducing the dimensionality of the search problem. A promising direction is to leverage data-driven methods to discover low-dimensional manifolds that capture the structure of task-relevant motions, enabling the controller to allocate samples along learned motion manifolds rather than uniformly across the raw actuation space. While cost-based weighting remains

effective regardless of the sampling source, establishing formal conditions under which latent-space sampling preserves the quality of the MPPI control estimate would provide a principled foundation for scaling MPPI to high-dimensional systems.

Third, the gap between simulation and real-world performance remains a challenge for all model-based control methods. Sensor noise, computational latency, actuator delays, and unmodeled physical effects collectively create mismatches between planned and executed trajectories. MPPI's multi-sample structure offers a natural avenue for addressing this gap, as techniques such as domain randomization can be directly embedded into the sampling process so that the cost-weighted average inherently favors control sequences that are robust across parameter variations. This capability is further strengthened by integration with physics simulators discussed in Section 4.1.4, where each rollout is propagated under physically realistic conditions including contact dynamics and frictional interactions. For computational latency, predictive state forwarding, where the initial condition is propagated by the expected delay before optimization begins, provides a practical compensation strategy. Fully exploiting these structural advantages to systematically close the sim-to-real gap remains a promising direction for future research. Ultimately, by addressing these open challenges, MPPI has the potential to evolve into a theoretically grounded framework for real-time stochastic optimal control in robotics.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used AI-assisted writing tools to organize the manuscript structure and improve language clarity. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

The simulation framework, experimental data, and implementation code supporting the findings of this study are publicly available at <https://anonymous.4open.science/r/mppi-framework/>.

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